4.00-hpi-german-cc-example

September 2, 2024

1 PRiSM Example: German Credit Card Data

This notebook demonstrates the PRiSM (Partial Responses in Structured Models) method, which is designed to transform black-box classifiers into inherently interpretable models without compromising predictive performance. This approach is particularly valuable in high-stakes applications where understanding the model's decision-making process is crucial.

The method addresses a key challenge in machine learning: the trade-off between model complexity and interpretability. It allows us to create models that are both accurate and transparent. In this example, we'll use a German credit card dataset to illustrate the PRiSM method. While this isn't a medical dataset, the principles demonstrated here are directly applicable to clinical data and decision support systems.

Here, the target outcome we are modelling is "Class label", which refers to:

- 0: a customer who **did not default** on their loan
- 1: a customer who **did default** on their loan

We would like to use the available features to model the likelihood of a customer defaulting, and understand how each feature contributes to that likelihood.

1.1 1. Setup and Imports

First, we need to import the necessary libraries and set up our environment.

[296]:
```
%load_ext autoreload
%autoreload 2

import numpy as np
import pandas as pd
import torch
import matplotlib.pyplot as plt
import seaborn as sns

from prism.config import RAW_DATA_DIR, MODELS_DIR
from prism.preprocessing import normalize, feature_summary,␣
 ↪plot_feature_histograms
from prism.maskedmlp import train_mlp_batched, train_maskedmlp
from prism.partial_responses import partial_responses
from prism.response_plot import plot_partial_responses
```

```
from prism.nomogram import nomogram, display_nomograms_side_by_side
from prism.lasso import lasso
from prism.device_tools import get_device, get_num_cpu_workers
from prism.metrics import evaluate_model_performance, compare_model_performance
```

```
The autoreload extension is already loaded. To reload it, use:
  %reload_ext autoreload
```

[297]:
```
%reload_ext autoreload
```

We set up our computational device (CPU or GPU if available) and set a random seed for reproducibility.

We'll use the 'lebesgue' method for partial responses, which is one of two methods (the other being 'dirac') for calculating partial responses. The Lebesgue method uses an average over the predicted surface, which can be more stable for some datasets, but more computationally intensive. If you're working with a bigger dataset, or low compute resources, starting with 'dirac' may be better.

[298]:
```
# Set device to 'cpu', or GPU ('cuda' for NVIDIA, 'mps' for Apple Silicon)
# By default, we select the best available device
device = get_device()
print(f"Using device: {device}")

# Set random seed for reproducibility
random_seed = 257
np.random.seed(random_seed)
torch.manual_seed(random_seed)

# Set method and other parameters
partial_response_method  = 'lebesgue'
SAVE_METRICS = True

# TODO: setup model saving.
SAVE_MODELS = False
```

```
Using device: cuda
```

For multithreading compute-intensive tasks, the recommended number of workers for GPU processing is 1, otherwise for CPU stick to the default (number of logical cores - 1). However, depending on your hardware, you may want to experiment.

[299]:
```
if device != 'cpu':
    max_workers = 1
else:
    max_workers = get_num_cpu_workers()
print(f"max_workers = {max_workers}")
```

```
max_workers = 1
```

1.2 2. Load and Preprocess Data

Next, we load the German credit card data and perform necessary preprocessing steps.

[300]:
```python
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
from sklearn.impute import SimpleImputer

def preprocess_german_credit_data(data, target_column, feature_columns,␣
↪test_size=0.2, random_state=257, shuffle=False, credit_scale=0.01):
    """
    Preprocess the German Credit Card dataset.

    Parameters:
    data (pd.DataFrame): The raw German Credit Card dataset
    target_column (str or int): The name or index of the target column
    feature_columns (list): List of column names or indices to use as features
    test_size (float): Proportion of the dataset to include in the test split
    random_state (int): Random state for reproducibility

    Returns:
    tuple: X_train, X_test, y_train, y_test (all as pd.DataFrame or pd.Series)
    """
    # Select features and target
    if isinstance(target_column, int):
        target_column = data.columns[target_column]

    if all(isinstance(col, int) for col in feature_columns):
        feature_columns = data.columns[feature_columns]

    X = data.copy()[feature_columns]
    y = data[target_column]

    # Scale the 'Credit amount' if it exists in the dataset
    if 'Credit amount' in data.columns:
        X['Credit amount'] = X['Credit amount'] * credit_scale
        X['Credit amount'] = X['Credit amount'].round()

    # Encode categorical variables
    label_encoder = LabelEncoder()
    for column in X.select_dtypes(include=['object']):
        X[column] = label_encoder.fit_transform(X[column].astype(str))

    # Impute missing values
    imputer = SimpleImputer(strategy='median')
    X = pd.DataFrame(imputer.fit_transform(X), columns=X.columns)

    # Split the data into training and testing sets
```

```
    X_train, X_test, y_train, y_test = train_test_split(X, y,␣
↪test_size=test_size,random_state=random_seed,shuffle=shuffle)

    # Ensure the target variable is binary
    y_train = (y_train == y_train.max()).astype(int)
    y_test = (y_test == y_test.max()).astype(int)

    return X_train, X_test, y_train, y_test
```

[301]:
```
# Load data
german_credit_data = pd.read_csv(RAW_DATA_DIR.joinpath('GermanCC_attributes.
↪csv'))

target_column = 'Class label'
feature_columns = [0, 1, 2, 4, 5, 6, 13, 19, 20, 21]  # List of column indices␣
↪to use as features

X_train, X_test, y_train, y_test = preprocess_german_credit_data(
    german_credit_data,
    target_column,
    feature_columns,
    credit_scale=0.01,
    test_size=0.3,
    random_state=random_seed
)

# Normalize the data to ensure all features have a median of zero
X_train_normalized, X_test_normalized = normalize(X_train, X_test)

# Convert to tensors for use with PyTorch
X_train_tensor = torch.tensor(X_train_normalized.values, dtype=torch.float32,␣
↪device=device)
X_test_tensor = torch.tensor(X_test_normalized.values, dtype=torch.float32,␣
↪device=device)
y_train_tensor = torch.tensor(y_train.values, dtype=torch.float32,␣
↪device=device)
y_test_tensor = torch.tensor(y_test.values, dtype=torch.float32, device=device)

feature_names = [
    'Account Status',
    'Duration (Months)',
    'Credit History',
    'Credit Amount',
    'Savings/Bonds',
    'Employment Length',
    'Other Plans',
    'Foreign Worker',
```

```
    'Unknown1',
    'Unknown2'
]
```

1.3 3. Data Overview

Before building our models, we'll get an overview of the data.

[302]:
```
print("Shape of training data:", X_train.shape)
print("Shape of testing data:", X_test.shape)
print("\nClass distribution in training set:")
print(y_train.value_counts(normalize=True))
print("\nClass distribution in test set:")
print(y_test.value_counts(normalize=True))
```

```
Shape of training data: (700, 10)
Shape of testing data: (300, 10)

Class distribution in training set:
Class label
0    0.704286
1    0.295714
Name: proportion, dtype: float64

Class distribution in test set:
Class label
0    0.69
1    0.31
Name: proportion, dtype: float64
```

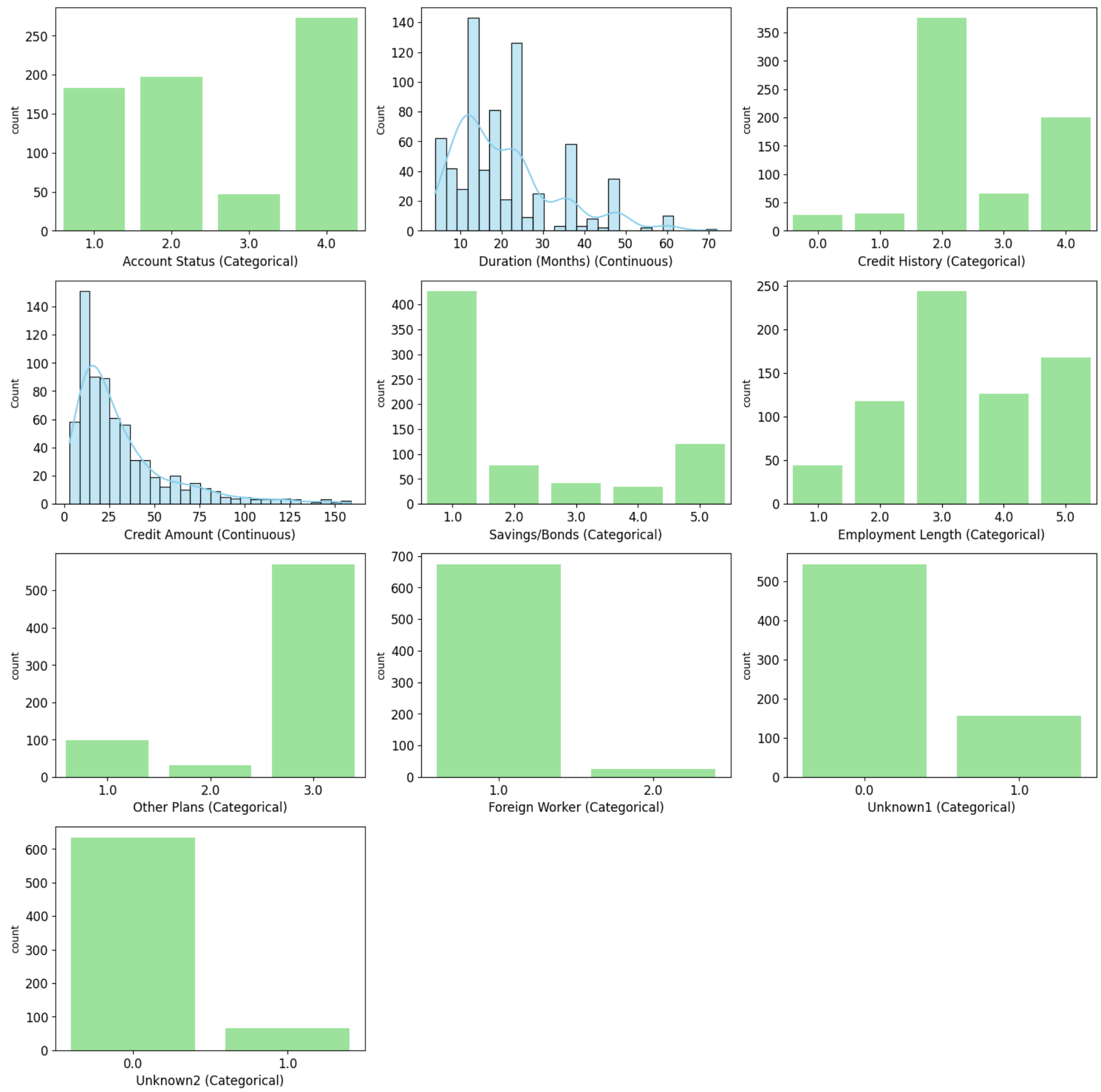
[303]:
```
feature_stats = feature_summary(X_train, categorical_threshold = 15)
print("Feature Statistics:")
print(feature_stats)
```

```
Feature Statistics:
                                   Data Type  Non-Null Count  Null Count  \
Status of existing checking account   float64             700           0
Duration in month                     float64             700           0
Credit history                        float64             700           0
Credit amount                         float64             700           0
Savings account/bonds                 float64             700           0
Period of present employment          float64             700           0
Other installment plans               float64             700           0
Foreign worker                        float64             700           0
Unknown1                              float64             700           0
unknown2                              float64             700           0

                                         Mean  Median   Std Dev   IQR  Min  \
Status of existing checking account   2.585714     2.0  1.244074  3.00  1.0
```

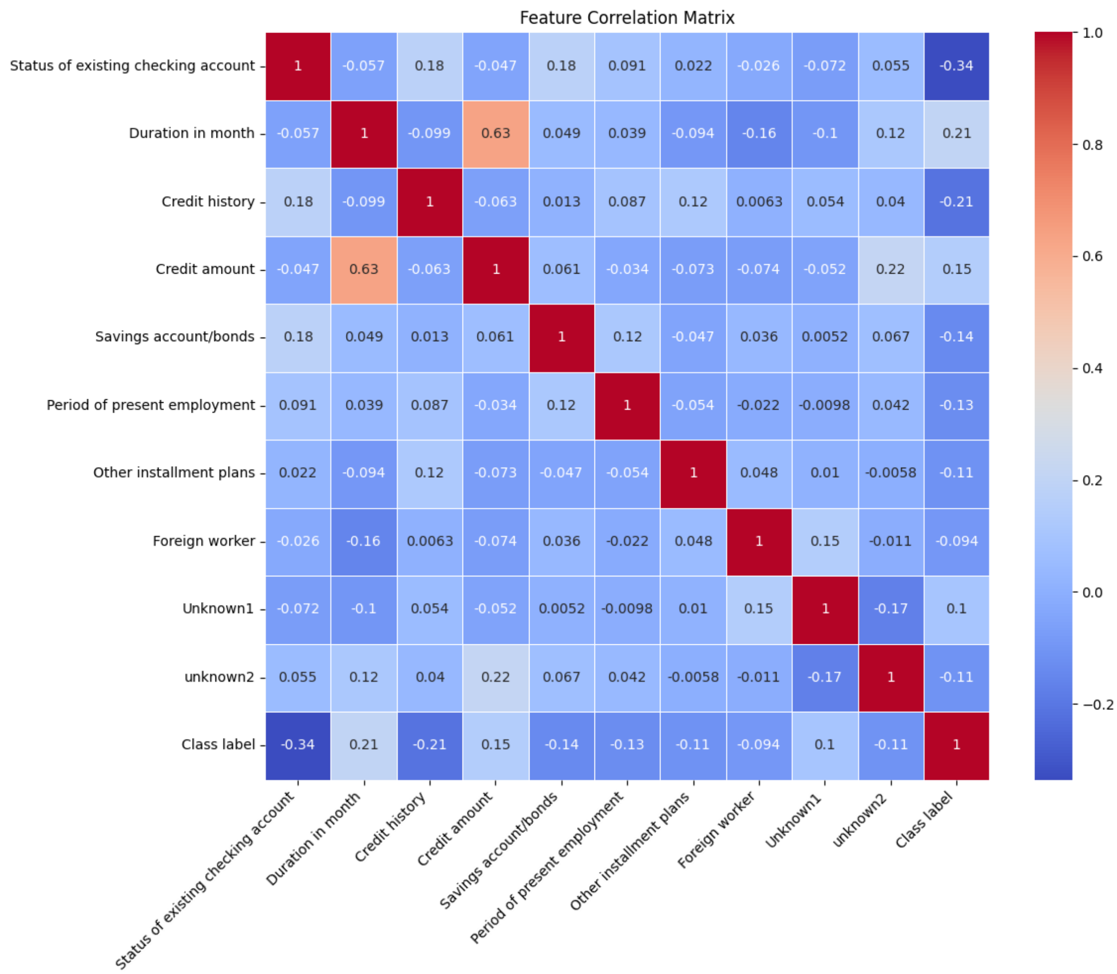
```
Duration in month                     20.652857    18.0  12.279683  12.00  4.0
Credit history                         2.542857     2.0   1.070955   2.00  0.0
Credit amount                         31.822857    23.0  27.278724  25.25  3.0
Savings account/bonds                  2.061429     1.0   1.551389   2.00  1.0
Period of present employment           3.365714     3.0   1.195112   1.00  1.0
Other installment plans                2.674286     3.0   0.707347   0.00  1.0
Foreign worker                         1.037143     1.0   0.189247   0.00  1.0
Unknown1                               0.224286     0.0   0.417409   0.00  0.0
unknown2                               0.092857     0.0   0.290440   0.00  0.0

                                        Max  Unique Values  Is Categorical
Status of existing checking account     4.0              4            True
Duration in month                      72.0             32           False
Credit history                          4.0              5            True
Credit amount                         159.0            112           False
Savings account/bonds                   5.0              5            True
Period of present employment            5.0              5            True
Other installment plans                 3.0              3            True
Foreign worker                          2.0              2            True
Unknown1                                1.0              2            True
unknown2                                1.0              2            True
```

[330]:

```
plot_feature_histograms(X_train,␣
    ↪feature_stats,figsize=(15,15),feature_names=[*feature_names,'Class label']);
```



[305]:
```python
# Correlation matrix
correlation_matrix = pd.concat([X_train, y_train], axis=1).corr()
plt.figure(figsize=(12, 10))
sns.heatmap(correlation_matrix, annot=True, cmap='coolwarm', linewidths=0.5)
plt.title('Feature Correlation Matrix')
plt.xticks(rotation=45, ha='right')
plt.tight_layout()
plt.show()
```



1.4 4. Train Initial Black Box Model (MLP)

We start by training a black box model, in this case, a Multi-Layer Perceptron (MLP) with 10 nodes in the hidden layer. This model serves as our baseline and represents the type of hard-to-interpret model that is often used in machine learning.

[306]:
```
mlp_hyperparameters = {
    'n_hidden': 10,
    'weight_decay': 0.0001,
    'lr': 0.001,
    'patience': 10,
    'tolerance': 0.00001,
    'batch_size': 32,
    'device': device,
    'seed': random_seed
}
```

```
blackbox_model = train_mlp_batched(X_train_normalized, y_train, X_test_normalized, y_test, **mlp_hyperparameters)
```

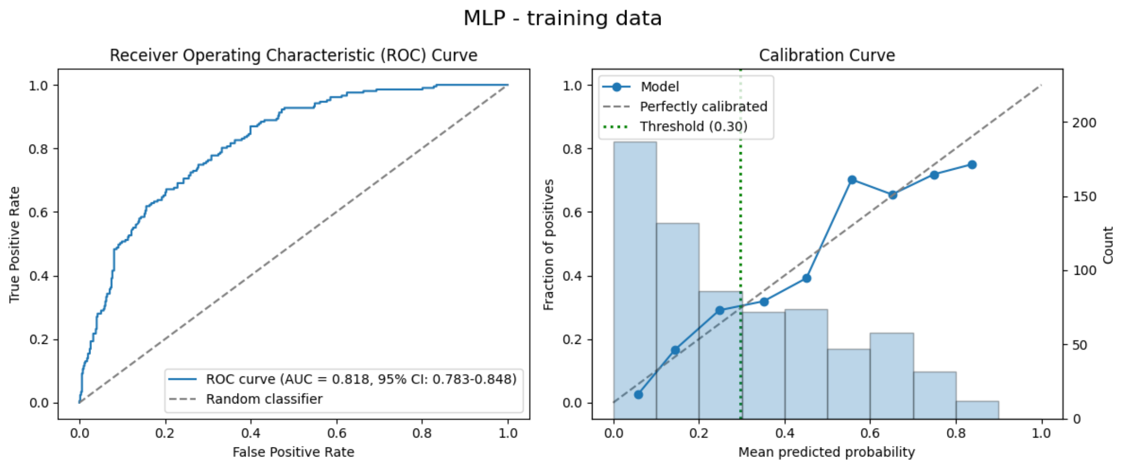
```
Epoch 0: Train loss 0.7623, Val loss 0.7541
Epoch 1: Train loss 0.7124, Val loss 0.7045
Epoch 2: Train loss 0.6719, Val loss 0.6644
Epoch 3: Train loss 0.6382, Val loss 0.6327
Epoch 4: Train loss 0.6130, Val loss 0.6061
Epoch 5: Train loss 0.5907, Val loss 0.5855
Epoch 6: Train loss 0.5744, Val loss 0.5692
Epoch 7: Train loss 0.5612, Val loss 0.5558
Epoch 8: Train loss 0.5499, Val loss 0.5451
Epoch 9: Train loss 0.5415, Val loss 0.5359
Epoch 10: Train loss 0.5342, Val loss 0.5282
Epoch 11: Train loss 0.5271, Val loss 0.5220
Epoch 12: Train loss 0.5226, Val loss 0.5164
Epoch 13: Train loss 0.5185, Val loss 0.5119
Epoch 14: Train loss 0.5133, Val loss 0.5073
Epoch 15: Train loss 0.5108, Val loss 0.5036
Epoch 16: Train loss 0.5062, Val loss 0.5005
Epoch 17: Train loss 0.5043, Val loss 0.4977
Epoch 18: Train loss 0.5014, Val loss 0.4953
Epoch 19: Train loss 0.4994, Val loss 0.4930
Epoch 20: Train loss 0.4980, Val loss 0.4913
Epoch 21: Train loss 0.4956, Val loss 0.4894
Epoch 22: Train loss 0.4945, Val loss 0.4878
Epoch 23: Train loss 0.4941, Val loss 0.4865
Epoch 24: Train loss 0.4918, Val loss 0.4852
Epoch 25: Train loss 0.4900, Val loss 0.4844
Epoch 26: Train loss 0.4884, Val loss 0.4831
Epoch 27: Train loss 0.4884, Val loss 0.4827
Epoch 28: Train loss 0.4879, Val loss 0.4814
Epoch 29: Train loss 0.4862, Val loss 0.4808
Epoch 30: Train loss 0.4855, Val loss 0.4802
Epoch 31: Train loss 0.4849, Val loss 0.4796
Epoch 32: Train loss 0.4841, Val loss 0.4792
Epoch 33: Train loss 0.4836, Val loss 0.4789
Epoch 34: Train loss 0.4833, Val loss 0.4785
Epoch 35: Train loss 0.4827, Val loss 0.4783
Epoch 36: Train loss 0.4821, Val loss 0.4778
Epoch 37: Train loss 0.4826, Val loss 0.4776
Epoch 38: Train loss 0.4816, Val loss 0.4773
Epoch 39: Train loss 0.4815, Val loss 0.4772
Epoch 40: Train loss 0.4812, Val loss 0.4768
Epoch 41: Train loss 0.4801, Val loss 0.4769
Epoch 42: Train loss 0.4800, Val loss 0.4767
Epoch 43: Train loss 0.4798, Val loss 0.4763
Epoch 44: Train loss 0.4796, Val loss 0.4764
```

```
Epoch 45: Train loss 0.4797, Val loss 0.4763
Epoch 46: Train loss 0.4794, Val loss 0.4762
Epoch 47: Train loss 0.4794, Val loss 0.4763
Epoch 48: Train loss 0.4787, Val loss 0.4762
Epoch 49: Train loss 0.4782, Val loss 0.4757
Epoch 50: Train loss 0.4777, Val loss 0.4762
Epoch 51: Train loss 0.4780, Val loss 0.4758
Epoch 52: Train loss 0.4769, Val loss 0.4758
Epoch 53: Train loss 0.4780, Val loss 0.4756
Epoch 54: Train loss 0.4779, Val loss 0.4757
Epoch 55: Train loss 0.4772, Val loss 0.4755
Epoch 56: Train loss 0.4765, Val loss 0.4754
Epoch 57: Train loss 0.4770, Val loss 0.4755
Epoch 58: Train loss 0.4761, Val loss 0.4755
Epoch 59: Train loss 0.4763, Val loss 0.4755
Epoch 60: Train loss 0.4760, Val loss 0.4754
Epoch 61: Train loss 0.4768, Val loss 0.4755
Epoch 62: Train loss 0.4751, Val loss 0.4752
Epoch 63: Train loss 0.4750, Val loss 0.4749
Epoch 64: Train loss 0.4771, Val loss 0.4754
Epoch 65: Train loss 0.4748, Val loss 0.4749
Epoch 66: Train loss 0.4750, Val loss 0.4748
Epoch 67: Train loss 0.4750, Val loss 0.4751
Epoch 68: Train loss 0.4748, Val loss 0.4748
Epoch 69: Train loss 0.4747, Val loss 0.4749
Epoch 70: Train loss 0.4737, Val loss 0.4750
Epoch 71: Train loss 0.4731, Val loss 0.4749
Epoch 72: Train loss 0.4738, Val loss 0.4750
Epoch 73: Train loss 0.4733, Val loss 0.4746
Epoch 74: Train loss 0.4734, Val loss 0.4747
Epoch 75: Train loss 0.4731, Val loss 0.4747
Epoch 76: Train loss 0.4730, Val loss 0.4748
Epoch 77: Train loss 0.4728, Val loss 0.4746
Epoch 78: Train loss 0.4724, Val loss 0.4748
Epoch 79: Train loss 0.4734, Val loss 0.4747
Epoch 80: Train loss 0.4729, Val loss 0.4748
Epoch 81: Train loss 0.4720, Val loss 0.4749
Epoch 82: Train loss 0.4717, Val loss 0.4746
Epoch 83: Train loss 0.4719, Val loss 0.4748
Epoch 84: Train loss 0.4708, Val loss 0.4744
Epoch 85: Train loss 0.4710, Val loss 0.4746
Epoch 86: Train loss 0.4717, Val loss 0.4743
Epoch 87: Train loss 0.4706, Val loss 0.4744
Epoch 88: Train loss 0.4703, Val loss 0.4745
Epoch 89: Train loss 0.4700, Val loss 0.4744
Epoch 90: Train loss 0.4708, Val loss 0.4744
Epoch 91: Train loss 0.4707, Val loss 0.4741
Epoch 92: Train loss 0.4695, Val loss 0.4745
```

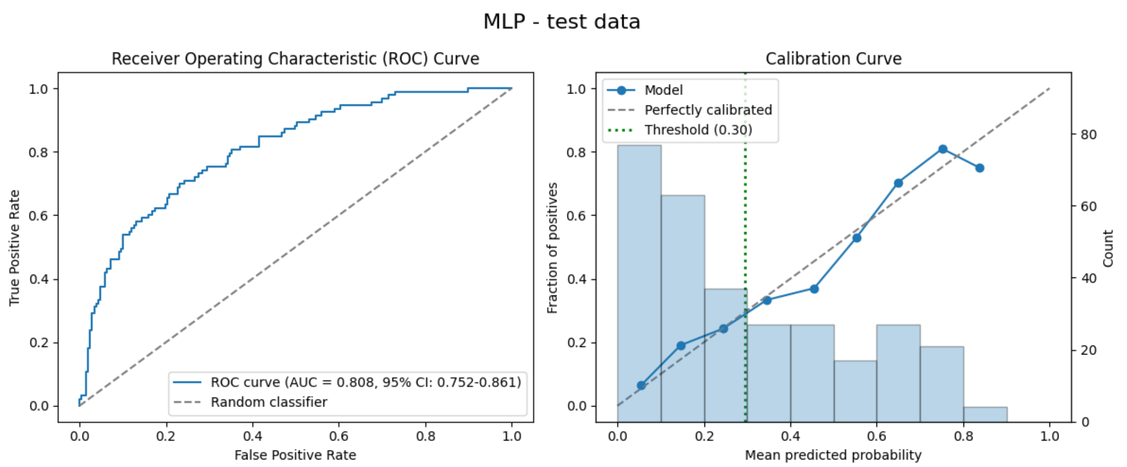
```
Epoch 93: Train loss 0.4700, Val loss 0.4743
Epoch 94: Train loss 0.4693, Val loss 0.4742
Epoch 95: Train loss 0.4688, Val loss 0.4745
Epoch 96: Train loss 0.4691, Val loss 0.4743
Epoch 97: Train loss 0.4686, Val loss 0.4743
Epoch 98: Train loss 0.4690, Val loss 0.4741
Epoch 99: Train loss 0.4681, Val loss 0.4741
Epoch 100: Train loss 0.4682, Val loss 0.4742
Epoch 101: Train loss 0.4668, Val loss 0.4743
Epoch 102: Train loss 0.4678, Val loss 0.4744
Epoch 103: Train loss 0.4671, Val loss 0.4738
Epoch 104: Train loss 0.4688, Val loss 0.4743
Epoch 105: Train loss 0.4670, Val loss 0.4745
Epoch 106: Train loss 0.4673, Val loss 0.4741
Epoch 107: Train loss 0.4673, Val loss 0.4741
Epoch 108: Train loss 0.4662, Val loss 0.4741
Epoch 109: Train loss 0.4667, Val loss 0.4739
Epoch 110: Train loss 0.4658, Val loss 0.4740
Epoch 111: Train loss 0.4656, Val loss 0.4743
Epoch 112: Train loss 0.4664, Val loss 0.4741
Stopping early at epoch 103
```

[308]:
```
y_pred_train_blackbox = blackbox_model.predict(X_train_tensor)
y_pred_test_blackbox = blackbox_model.predict(X_test_tensor)

# Evaluate MLP
results_mlp_train = evaluate_model_performance(y_train, y_pred_train_blackbox,
 ↪y_train, title="MLP - training data")
results_mlp_test = evaluate_model_performance(y_test, y_pred_test_blackbox,
 ↪y_train, title="MLP - test data")
```

```
---- Model Performance Metrics MLP - training data ----
Threshold (prevalence) set to: 0.296
AUROC:          0.818 (95% CI: 0.783-0.848) (Area Under the Receiver Operating
Characteristic curve)
Accuracy:       0.723 (Proportion of correct predictions)
Precision:      0.522 (Proportion of true positives among positive predictions,
aka PPV)
Recall:         0.754 (Proportion of true positives among actual positives, aka
sensitivity)
F1 Score:       0.617 (Harmonic mean of precision and recall)
--------------------------------------------
```




```
---- Model Performance Metrics MLP - test data ----
Threshold (prevalence) set to: 0.296
AUROC:          0.808 (95% CI: 0.752-0.861) (Area Under the Receiver Operating
Characteristic curve)
Accuracy:       0.723 (Proportion of correct predictions)
Precision:      0.540 (Proportion of true positives among positive predictions,
aka PPV)
Recall:         0.720 (Proportion of true positives among actual positives, aka
sensitivity)
F1 Score:       0.618 (Harmonic mean of precision and recall)
--------------------------------------------
```



1.5 5. Calculate Partial Responses

Now we calculate partial responses for our black box model. This is a key step in the PRiSM method. Partial responses help us understand how individual features or pairs of features contribute to the model's predictions, making the black box model more interpretable.

[309]:
```
partial_responses_params = {
    'x_train': X_train_tensor,
    'method': partial_response_method,
    'device': device,
    'batch_size': 512,
    'max_workers': max_workers
}

partial_responses_train = partial_responses(X_train_tensor, blackbox_model,␣
 ↪**partial_responses_params)
partial_responses_test = partial_responses(X_test_tensor, blackbox_model,␣
 ↪**partial_responses_params)
```

```
Main compute device: cuda
Max threads: 1
Batch size: 512
Univariate 0,   (cuda)
Univariate 1,   (cuda)
Univariate 2,   (cuda)
Univariate 3,   (cuda)
Univariate 4,   (cuda)
Univariate 5,   (cuda)
Univariate 6,   (cuda)
Univariate 7,   (cuda)
Univariate 8,   (cuda)
Univariate 9,   (cuda)
Bivariate 0,1,  (cuda)
Bivariate 0,2,  (cuda)
Bivariate 0,3,  (cuda)
Bivariate 0,4,  (cuda)
Bivariate 0,5,  (cuda)
Bivariate 0,6,  (cuda)
Bivariate 0,7,  (cuda)
Bivariate 0,8,  (cuda)
Bivariate 0,9,  (cuda)
Bivariate 1,2,  (cuda)
Bivariate 1,3,  (cuda)
Bivariate 1,4,  (cuda)
Bivariate 1,5,  (cuda)
Bivariate 1,6,  (cuda)
Bivariate 1,7,  (cuda)
Bivariate 1,8,  (cuda)
```

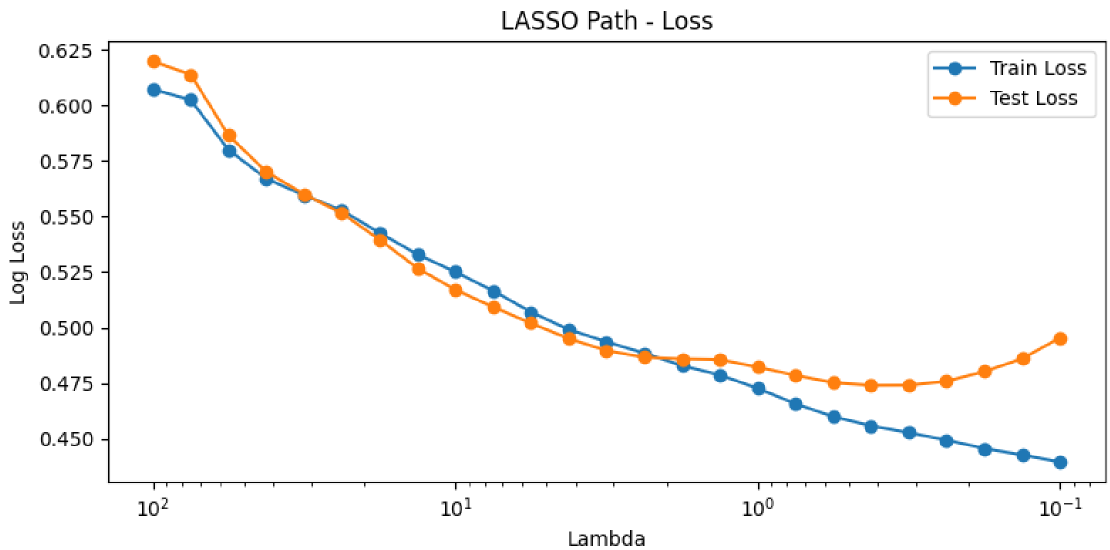
```
Bivariate 1,9,  (cuda)
Bivariate 2,3,  (cuda)
Bivariate 2,4,  (cuda)
Bivariate 2,5,  (cuda)
Bivariate 2,6,  (cuda)
Bivariate 2,7,  (cuda)
Bivariate 2,8,  (cuda)
Bivariate 2,9,  (cuda)
Bivariate 3,4,  (cuda)
Bivariate 3,5,  (cuda)
Bivariate 3,6,  (cuda)
Bivariate 3,7,  (cuda)
Bivariate 3,8,  (cuda)
Bivariate 3,9,  (cuda)
Bivariate 4,5,  (cuda)
Bivariate 4,6,  (cuda)
Bivariate 4,7,  (cuda)
Bivariate 4,8,  (cuda)
Bivariate 4,9,  (cuda)
Bivariate 5,6,  (cuda)
Bivariate 5,7,  (cuda)
Bivariate 5,8,  (cuda)
Bivariate 5,9,  (cuda)
Bivariate 6,7,  (cuda)
Bivariate 6,8,  (cuda)
Bivariate 6,9,  (cuda)
Bivariate 7,8,  (cuda)
Bivariate 7,9,  (cuda)
Bivariate 8,9,  (cuda)
Main compute device: cuda
Max threads: 1
Batch size: 512
Univariate 0,   (cuda)
Univariate 1,   (cuda)
Univariate 2,   (cuda)
Univariate 3,   (cuda)
Univariate 4,   (cuda)
Univariate 5,   (cuda)
Univariate 6,   (cuda)
Univariate 7,   (cuda)
Univariate 8,   (cuda)
Univariate 9,   (cuda)
Bivariate 0,1,  (cuda)
Bivariate 0,2,  (cuda)
Bivariate 0,3,  (cuda)
Bivariate 0,4,  (cuda)
Bivariate 0,5,  (cuda)
Bivariate 0,6,  (cuda)
```

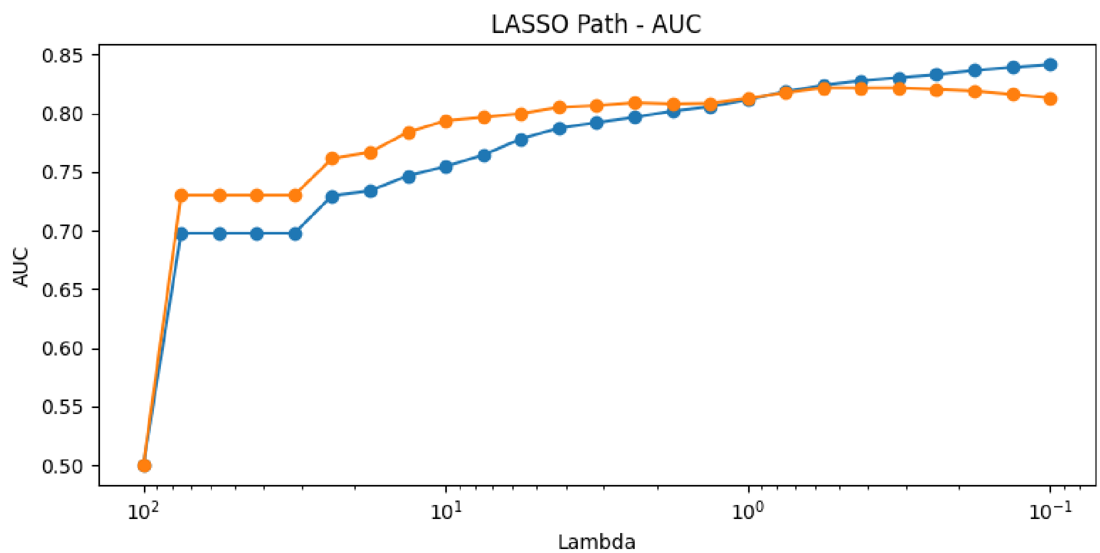
```
Bivariate 0,7,  (cuda)
Bivariate 0,8,  (cuda)
Bivariate 0,9,  (cuda)
Bivariate 1,2,  (cuda)
Bivariate 1,3,  (cuda)
Bivariate 1,4,  (cuda)
Bivariate 1,5,  (cuda)
Bivariate 1,6,  (cuda)
Bivariate 1,7,  (cuda)
Bivariate 1,8,  (cuda)
Bivariate 1,9,  (cuda)
Bivariate 2,3,  (cuda)
Bivariate 2,4,  (cuda)
Bivariate 2,5,  (cuda)
Bivariate 2,6,  (cuda)
Bivariate 2,7,  (cuda)
Bivariate 2,8,  (cuda)
Bivariate 2,9,  (cuda)
Bivariate 3,4,  (cuda)
Bivariate 3,5,  (cuda)
Bivariate 3,6,  (cuda)
Bivariate 3,7,  (cuda)
Bivariate 3,8,  (cuda)
Bivariate 3,9,  (cuda)
Bivariate 4,5,  (cuda)
Bivariate 4,6,  (cuda)
Bivariate 4,7,  (cuda)
Bivariate 4,8,  (cuda)
Bivariate 4,9,  (cuda)
Bivariate 5,6,  (cuda)
Bivariate 5,7,  (cuda)
Bivariate 5,8,  (cuda)
Bivariate 5,9,  (cuda)
Bivariate 6,7,  (cuda)
Bivariate 6,8,  (cuda)
Bivariate 6,9,  (cuda)
Bivariate 7,8,  (cuda)
Bivariate 7,9,  (cuda)
Bivariate 8,9,  (cuda)
```

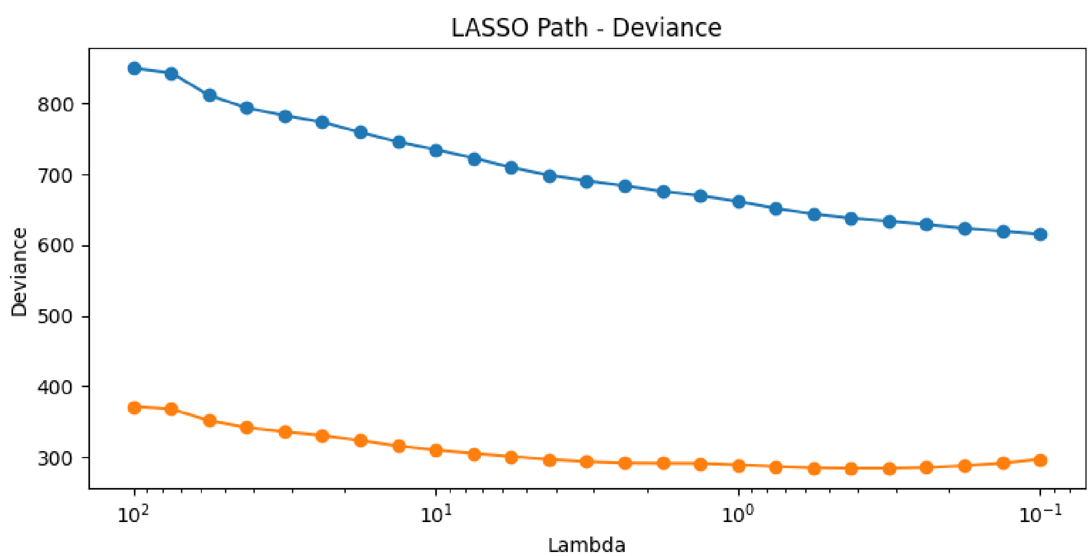
1.6 6. Perform LASSO on Partial Responses

We use LASSO (Least Absolute Shrinkage and Selection Operator) to select the most consequential partial responses. This step helps us identify which features or feature interactions contribute most to the model's predictions. A range of regularization strengths (lambda) are used, and for each one a logistic regression model with LASSO regularization is trained and evaluated. Based on the results, we can choose the most suitable lambda, balancing model performance against the total number of selected features.

[310]:
```
lasso_results = lasso(
    partial_responses_train,
    partial_responses_test,
    y_train,
    y_test,
    feature_names=feature_names,
    nlambda=25,
    min_lambda=0.1,
    max_lambda=100,
    batch_size=2,
    seed=random_seed
)
```


LASSO Path - Loss
Train Loss
Test Loss
Log Loss
Lambda
0.625
0.600
0.575
0.550
0.525
0.500
0.475
0.450
10^2
10^1
10^0
10^{-1}

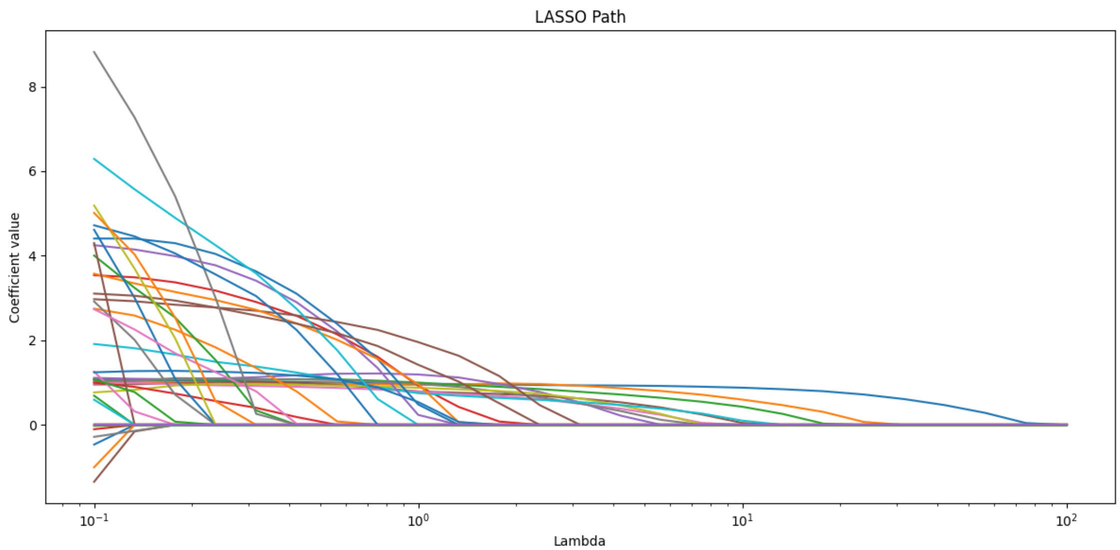

LASSO Path - AUC
AUC
Lambda
0.85
0.80
0.75
0.70
0.65
0.60
0.55
0.50
10^2
10^1
10^0
10^{-1}


LASSO Path - Deviance
Deviance
Lambda
800
700
600
500
400
300
10^2
10^1
10^0
10^{-1}

<IPython.core.display.HTML object>

All fits converged successfully.

[333]:
```
lasso_results.plot_lambda_path()
```



[335]:
```
lasso_results.select_lambda_max_test_auc()

# Alternatively, manually select lambda as needed
# lasso_results.select_lambda(18)
```

```
Selected lambda index: 18
Selected lambda value: 0.5623
Corresponding test AUC: 0.8217
```

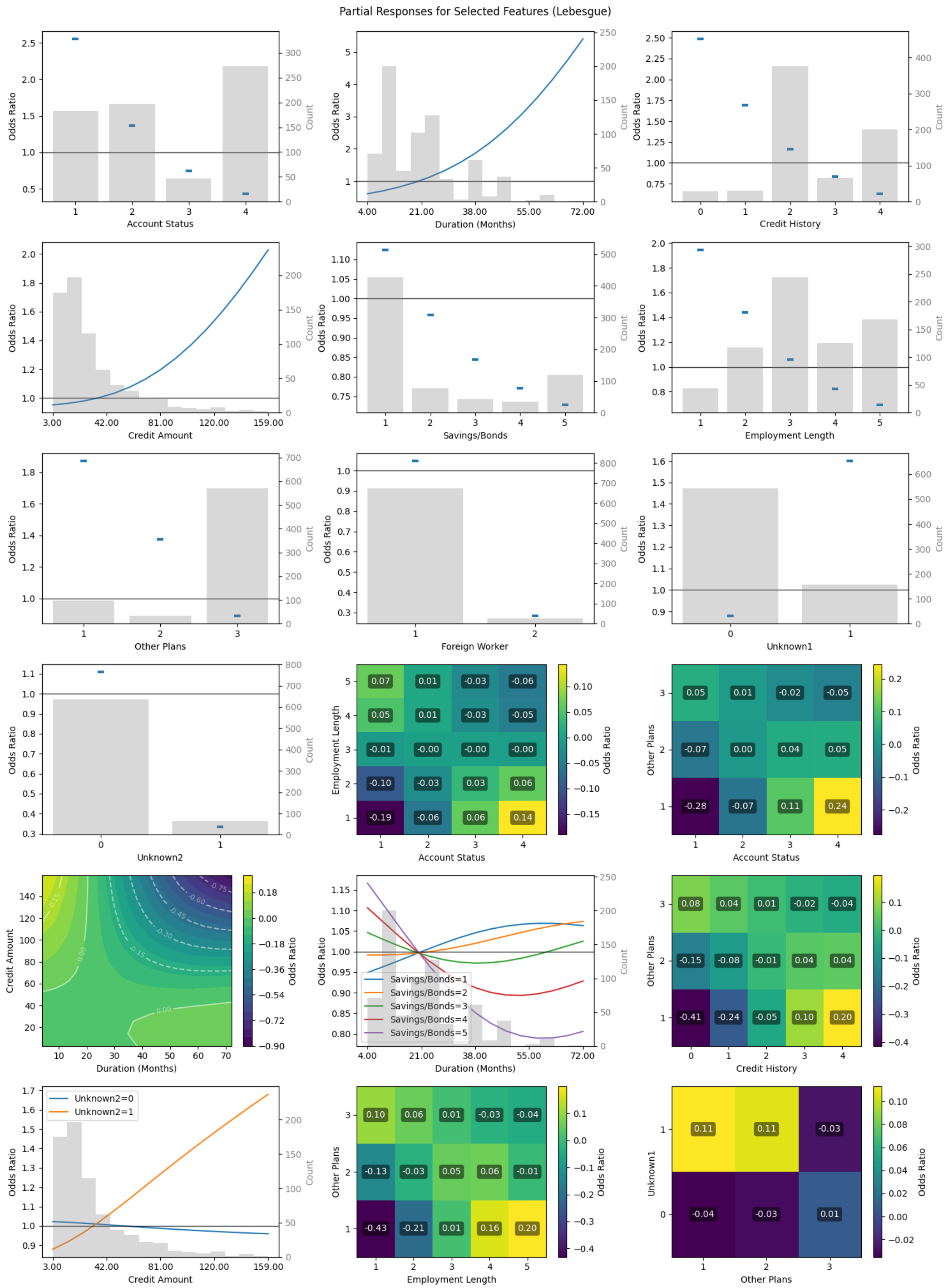
[335]: 18

1.7 7. View Selected Partial Responses and Generate Nomogram

First, we visualize the contribution of each partial response selected with LASSO to the target value.

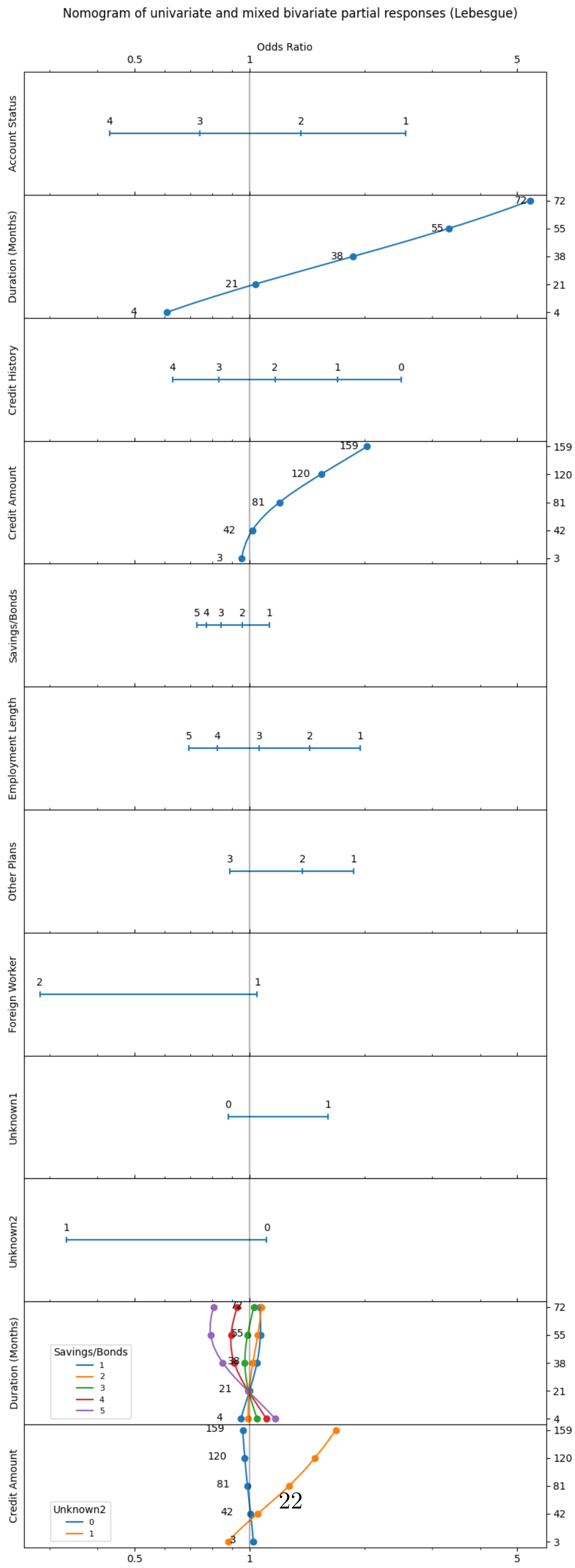
[316]:
```
fig_blackbox = plot_partial_responses(
    lasso_results,
    X_train_tensor,
    X_train.median().values,
    X_train.std().values,
    blackbox_model,
```

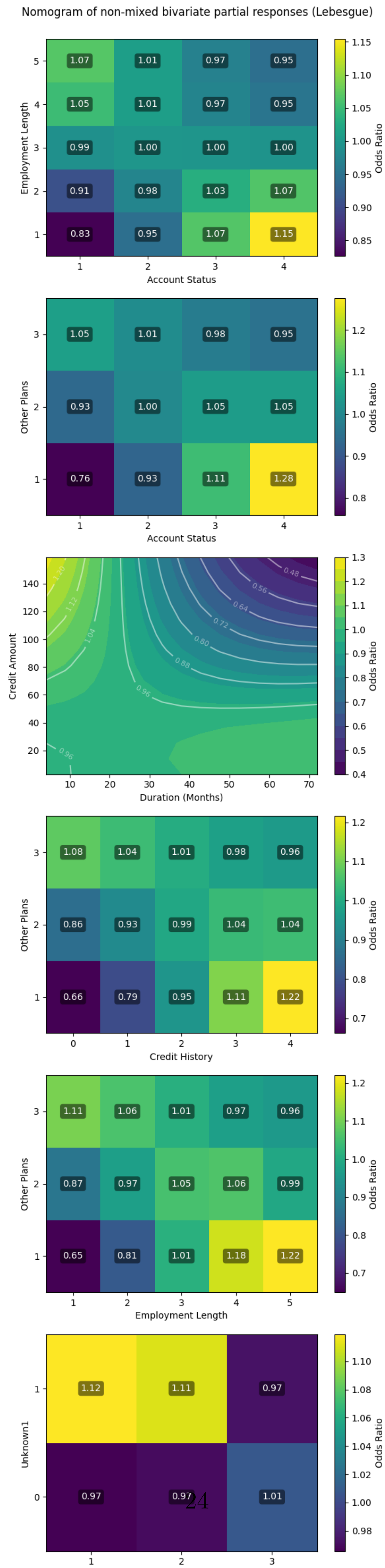
```
    n_steps=15,
    sd_scale=2,
    method=partial_response_method,
    device=device,
    categorical_threshold=15,
    subtract_univariate=True,
    figsize=(15,20),
    show_fig=True,
    return_fig=True,
    use_odds_ratio=True
)
```



We can also visualize the contribution of each partial response using a nomogram. Nomograms are particularly useful in clinical settings as they provide an intuitive visual representation of a model's decision-making process.

[317]:
```
nomogram_params = {
    'n_steps' : 15,
    'sd_scale' : 2,
    'method' : partial_response_method,
    'device' : device,
    'categorical_threshold' : 15,
    'subtract_univariate' : True,
    'return_fig' : True,
    'use_odds_ratio' : True,
    'save_csv' : SAVE_METRICS,
    'csv_comment' : "MLP German CC data"
}

nomogram_results = nomogram(
    lasso_results,
    X_train_tensor,
    X_train.median().values,
    X_train.std().values,
    blackbox_model,
    **nomogram_params
)
nomogram_main_mlp = nomogram_results[6]
nomogram_non_mixed_mlp = nomogram_results[7]
```


Nomogram of univariate and mixed bivariate partial responses (Lebesgue)
Odds Ratio
0.5
1
5
Account Status
Duration (Months)
Credit History
Credit Amount
Savings/Bonds
Employment Length
Other Plans
Foreign Worker
Unknown1
Unknown2
Duration (Months)
Savings/Bonds
Credit Amount
Unknown2
72
55
38
21
4
159
120
81
42
3


Nomogram of non-mixed bivariate partial responses (Lebesgue)
Employment Length
Account Status
Odds Ratio
1.07 1.01 0.97 0.95
1.05 1.01 0.97 0.95
0.99 1.00 1.00 1.00
0.91 0.98 1.03 1.07
0.83 0.95 1.07 1.15
Other Plans
Account Status
Odds Ratio
1.05 1.01 0.98 0.95
0.93 1.00 1.05 1.05
0.76 0.93 1.11 1.28
Credit Amount
Duration (Months)
Odds Ratio
Other Plans
Credit History
Odds Ratio
1.08 1.04 1.01 0.98 0.96
0.86 0.93 0.99 1.04 1.04
0.66 0.79 0.95 1.11 1.22
Other Plans
Employment Length
Odds Ratio
1.11 1.06 1.01 0.97 0.96
0.87 0.97 1.05 1.06 0.99
0.65 0.81 1.01 1.18 1.22
Unknown1
Other Plans
Odds Ratio
1.12 1.11 0.97
0.97 0.97 1.01

```
Univariate nomogram data saved to C:\Users\localuser\PRiSM\prism_github\models\n
omogram_20240830_1524_univariate.csv
Bivariate nomogram data saved to C:\Users\localuser\PRiSM\prism_github\models\no
mogram_20240830_1524_bivariate.csv
```

1.8 8. Train Partial Response Network (PRN)

Now we train a Partial Response Network based on the selected features. The PRN is a type of neural network that incorporates the interpretability of partial responses while maintaining the predictive power of the original black box model.

From the results of the lasso feature selection, we create an input mask for the PRN. The mask is a binary matrix that determines which input features connect to which hidden nodes (subnets) in the network. This enforces the network structure based on selected features.

The PRN uses a subnet architecture, where groups of hidden nodes are dedicated to specific features or feature interactions:

- Univariate subnets: Handle individual feature effects
- Bivariate subnets: Capture interactions between two features
- The number of nodes per subnet is controlled by the subnet_nodes parameter in `train_maskedmlp` (default 5)

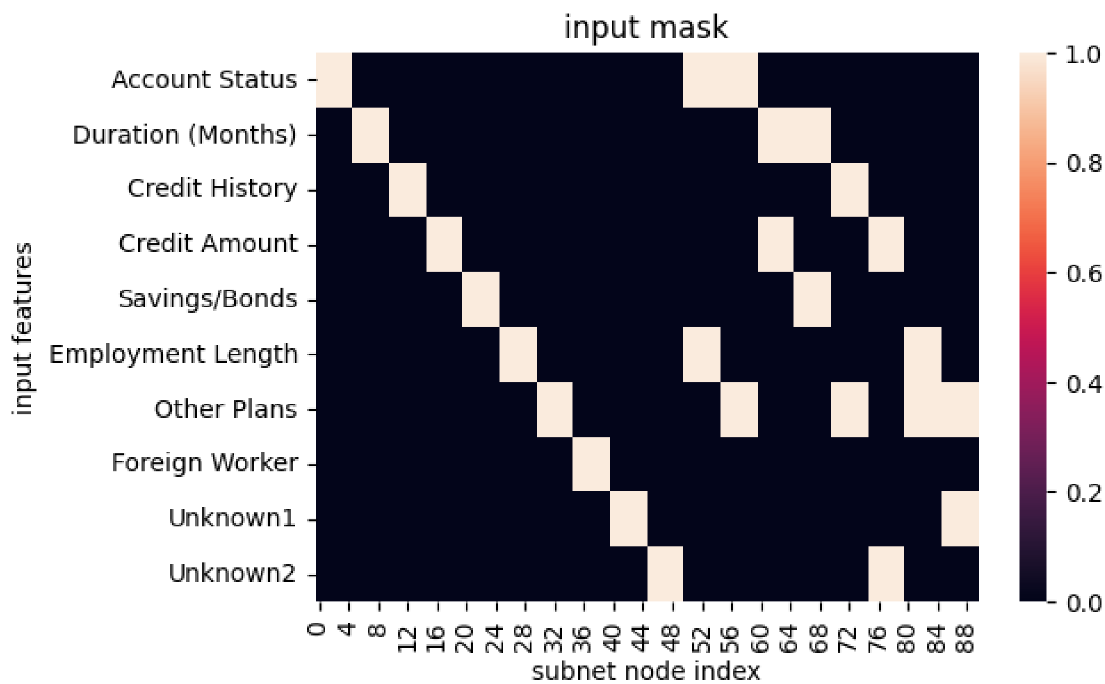
This gives us a resulting network structure as follows

- Input layer: Original features
- Hidden layer: Structured into subnets based on the mask
- Output layer: Single node for binary classification

The resulting PRN maintains interpretability because each subnet corresponds to a specific feature or interaction. Therefore, contribution of each feature/interaction can be analyzed separately, and the overall prediction is a sum of these interpretable components.

[318]:
```
mask, n_features = lasso_results.get_mask()
```

```
Selected features: ['Account Status', 'Duration (Months)', 'Credit History',
'Credit Amount', 'Savings/Bonds', 'Employment Length', 'Other Plans', 'Foreign
Worker', 'Unknown1', 'Unknown2', 'Account Status : Employment Length', 'Account
Status : Other Plans', 'Duration (Months) : Credit Amount', 'Duration (Months) :
Savings/Bonds', 'Credit History : Other Plans', 'Credit Amount : Unknown2',
'Employment Length : Other Plans', 'Other Plans : Unknown1']
```



[319]:

```
prn_hyperparameters = {
    'n_hidden': n_features,
    'mask': mask,
    'subnet_nodes': 5,
    'iter': 10000,
    'lr': 0.001,
    'weight_decay': 0.00001,
    'tolerance': 0.00001,
    'patience': 100,
    'device': device,
    'seed': random_seed
}

partial_response_network = train_maskedmlp(X_train_normalized, y_train,␣
 ↪X_test_normalized, y_test, **prn_hyperparameters)
```

```
Epoch 0, Training loss 0.6634210348129272, Test loss 0.6679077744483948
Epoch 1, Training loss 0.6700581312179565, Test loss 0.6635481715202332
Epoch 2, Training loss 0.6656980514526367, Test loss 0.6591731905937195
Epoch 3, Training loss 0.661327600479126, Test loss 0.6548444032669067
Epoch 4, Training loss 0.6570026278495789, Test loss 0.6505861282348633
Epoch 5, Training loss 0.6527457237243652, Test loss 0.6464112401008606
Epoch 6, Training loss 0.6485685706138611, Test loss 0.6423269510269165
Epoch 7, Training loss 0.6444780826568604, Test loss 0.638338029384613
Epoch 8, Training loss 0.6404786705970764, Test loss 0.634447455406189
```

```
Epoch 9, Training loss 0.636573314666748, Test loss 0.6306573748588562
Epoch 10, Training loss 0.6327640414237976, Test loss 0.6269692182540894
Epoch 11, Training loss 0.6290522813796997, Test loss 0.6233837604522705
Epoch 12, Training loss 0.6254389882087708, Test loss 0.6199013590812683
Epoch 13, Training loss 0.6219245195388794, Test loss 0.6165218949317932
Epoch 14, Training loss 0.61850905418396, Test loss 0.6132451295852661
Epoch 15, Training loss 0.6151924729347229, Test loss 0.6100702881813049
Epoch 16, Training loss 0.6119742393493652, Test loss 0.6069965362548828
Epoch 17, Training loss 0.6088537573814392, Test loss 0.604022741317749
Epoch 18, Training loss 0.6058301329612732, Test loss 0.6011475920677185
Epoch 19, Training loss 0.6029022932052612, Test loss 0.5983695387840271
Epoch 20, Training loss 0.6000691652297974, Test loss 0.5956868529319763
Epoch 21, Training loss 0.5973291993141174, Test loss 0.5930978655815125
Epoch 22, Training loss 0.5946810245513916, Test loss 0.5906006097793579
Epoch 23, Training loss 0.5921229124069214, Test loss 0.5881927609443665
Epoch 24, Training loss 0.5896532535552979, Test loss 0.5858724117279053
Epoch 25, Training loss 0.5872699618339539, Test loss 0.583636999130249
Epoch 26, Training loss 0.5849712491035461, Test loss 0.5814842581748962
Epoch 27, Training loss 0.5827550292015076, Test loss 0.5794118642807007
Epoch 28, Training loss 0.5806190371513367, Test loss 0.5774170160293579
Epoch 29, Training loss 0.5785611867904663, Test loss 0.5754971504211426
Epoch 30, Training loss 0.5765790343284607, Test loss 0.5736498236656189
Epoch 31, Training loss 0.5746703147888184, Test loss 0.5718721151351929
Epoch 32, Training loss 0.5728325843811035, Test loss 0.5701615810394287
Epoch 33, Training loss 0.5710633993148804, Test loss 0.5685152411460876
Epoch 34, Training loss 0.5693602561950684, Test loss 0.5669305920600891
Epoch 35, Training loss 0.5677206516265869, Test loss 0.5654048919677734
Epoch 36, Training loss 0.5661421418190002, Test loss 0.5639353394508362
Epoch 37, Training loss 0.564622163772583, Test loss 0.562519371509552
Epoch 38, Training loss 0.5631582140922546, Test loss 0.5611544251441956
Epoch 39, Training loss 0.5617478489875793, Test loss 0.5598377585411072
Epoch 40, Training loss 0.5603885650634766, Test loss 0.5585669875144958
Epoch 41, Training loss 0.5590780377388, Test loss 0.557339608669281
Epoch 42, Training loss 0.5578137636184692, Test loss 0.5561530590057373
Epoch 43, Training loss 0.5565935373306274, Test loss 0.5550052523612976
Epoch 44, Training loss 0.5554150342941284, Test loss 0.5538939237594604
Epoch 45, Training loss 0.5542762279510498, Test loss 0.5528167486190796
Epoch 46, Training loss 0.5531748533248901, Test loss 0.551771879196167
Epoch 47, Training loss 0.5521089434623718, Test loss 0.5507571697235107
Epoch 48, Training loss 0.5510765910148621, Test loss 0.5497709512710571
Epoch 49, Training loss 0.5500758290290833, Test loss 0.5488113164901733
Epoch 50, Training loss 0.5491049885749817, Test loss 0.5478765964508057
Epoch 51, Training loss 0.5481624007225037, Test loss 0.5469653606414795
Epoch 52, Training loss 0.5472463369369507, Test loss 0.5460760593414307
Epoch 53, Training loss 0.5463554859161377, Test loss 0.5452073216438293
Epoch 54, Training loss 0.5454883575439453, Test loss 0.5443578958511353
Epoch 55, Training loss 0.5446435213088989, Test loss 0.5435266494750977
Epoch 56, Training loss 0.5438199043273926, Test loss 0.542712390422821
```

```
Epoch 57, Training loss 0.5430163741111755, Test loss 0.5419142246246338
Epoch 58, Training loss 0.5422316193580627, Test loss 0.5411311984062195
Epoch 59, Training loss 0.5414648652076721, Test loss 0.540362536907196
Epoch 60, Training loss 0.5407150983810425, Test loss 0.5396073460578918
Epoch 61, Training loss 0.5399814248085022, Test loss 0.5388650894165039
Epoch 62, Training loss 0.5392631888389587, Test loss 0.5381349921226501
Epoch 63, Training loss 0.5385594964027405, Test loss 0.5374165773391724
Epoch 64, Training loss 0.5378697514533997, Test loss 0.5367093682289124
Epoch 65, Training loss 0.5371933579444885, Test loss 0.5360127687454224
Epoch 66, Training loss 0.5365296602249146, Test loss 0.5353266000747681
Epoch 67, Training loss 0.5358783006668091, Test loss 0.5346502661705017
Epoch 68, Training loss 0.5352386236190796, Test loss 0.533983588218689
Epoch 69, Training loss 0.5346103310585022, Test loss 0.5333262085914612
Epoch 70, Training loss 0.5339928865432739, Test loss 0.5326778888702393
Epoch 71, Training loss 0.5333860516548157, Test loss 0.5320383906364441
Epoch 72, Training loss 0.532789409160614, Test loss 0.5314075946807861
Epoch 73, Training loss 0.5322025418281555, Test loss 0.5307852029800415
Epoch 74, Training loss 0.5316253900527954, Test loss 0.5301710963249207
Epoch 75, Training loss 0.5310573577880859, Test loss 0.529565155506134
Epoch 76, Training loss 0.5304985046386719, Test loss 0.5289672017097473
Epoch 77, Training loss 0.5299485325813293, Test loss 0.528377115726471
Epoch 78, Training loss 0.5294069647789001, Test loss 0.5277948379516602
Epoch 79, Training loss 0.528873860836029, Test loss 0.5272201895713806
Epoch 80, Training loss 0.5283489227294922, Test loss 0.5266531705856323
Epoch 81, Training loss 0.5278318524360657, Test loss 0.526093602180481
Epoch 82, Training loss 0.5273225903511047, Test loss 0.5255414247512817
Epoch 83, Training loss 0.526820957660675, Test loss 0.5249965786933899
Epoch 84, Training loss 0.5263267159461975, Test loss 0.5244590044021606
Epoch 85, Training loss 0.5258396863937378, Test loss 0.5239284634590149
Epoch 86, Training loss 0.5253597497940063, Test loss 0.5234049558639526
Epoch 87, Training loss 0.5248867273330688, Test loss 0.5228884816169739
Epoch 88, Training loss 0.524420440196991, Test loss 0.5223789215087891
Epoch 89, Training loss 0.5239607691764832, Test loss 0.5218760371208191
Epoch 90, Training loss 0.5235075354576111, Test loss 0.5213798880577087
Epoch 91, Training loss 0.5230606198310852, Test loss 0.5208904147148132
Epoch 92, Training loss 0.5226197838783264, Test loss 0.5204074382781982
Epoch 93, Training loss 0.5221850275993347, Test loss 0.5199308395385742
Epoch 94, Training loss 0.5217560529708862, Test loss 0.5194606781005859
Epoch 95, Training loss 0.5213328003883362, Test loss 0.5189965963363647
Epoch 96, Training loss 0.520915150642395, Test loss 0.5185387134552002
Epoch 97, Training loss 0.5205029249191284, Test loss 0.5180869698524475
Epoch 98, Training loss 0.5200960040092468, Test loss 0.5176411271095276
Epoch 99, Training loss 0.5196942687034607, Test loss 0.5172010660171509
Epoch 100, Training loss 0.5192975401878357, Test loss 0.5167667865753174
Epoch 101, Training loss 0.518905758857727, Test loss 0.5163381099700928
Epoch 102, Training loss 0.5185188055038452, Test loss 0.515915036201477
Epoch 103, Training loss 0.5181366205215454, Test loss 0.5154973864555359
Epoch 104, Training loss 0.517758846282959, Test loss 0.5150851011276245
```

```
Epoch 105, Training loss 0.5173856616020203, Test loss 0.5146780610084534
Epoch 106, Training loss 0.5170167088508606, Test loss 0.5142761468887329
Epoch 107, Training loss 0.5166521072387695, Test loss 0.5138792991638184
Epoch 108, Training loss 0.5162916779518127, Test loss 0.5134873986244202
Epoch 109, Training loss 0.5159352421760559, Test loss 0.5131003856658936
Epoch 110, Training loss 0.5155828595161438, Test loss 0.512718141078949
Epoch 111, Training loss 0.5152342915534973, Test loss 0.5123404860496521
Epoch 112, Training loss 0.5148895382881165, Test loss 0.5119674801826477
Epoch 113, Training loss 0.5145484805107117, Test loss 0.5115990042686462
Epoch 114, Training loss 0.514211118221283, Test loss 0.5112348794937134
Epoch 115, Training loss 0.513877272605896, Test loss 0.5108750462532043
Epoch 116, Training loss 0.513546884059906, Test loss 0.5105194449424744
Epoch 117, Training loss 0.5132198929786682, Test loss 0.5101680159568787
Epoch 118, Training loss 0.5128962993621826, Test loss 0.5098206400871277
Epoch 119, Training loss 0.5125759243965149, Test loss 0.5094772577285767
Epoch 120, Training loss 0.5122588276863098, Test loss 0.5091378092765808
Epoch 121, Training loss 0.5119449496269226, Test loss 0.5088021755218506
Epoch 122, Training loss 0.5116340517997742, Test loss 0.5084702968597412
Epoch 123, Training loss 0.511326253414154, Test loss 0.5081421136856079
Epoch 124, Training loss 0.5110214352607727, Test loss 0.5078175067901611
Epoch 125, Training loss 0.5107195973396301, Test loss 0.5074964761734009
Epoch 126, Training loss 0.5104206800460815, Test loss 0.5071789622306824
Epoch 127, Training loss 0.5101245641708374, Test loss 0.5068649053573608
Epoch 128, Training loss 0.5098313689231873, Test loss 0.506554126739502
Epoch 129, Training loss 0.509540855884552, Test loss 0.5062466263771057
Epoch 130, Training loss 0.5092530846595764, Test loss 0.5059425234794617
Epoch 131, Training loss 0.5089679956436157, Test loss 0.505641520023346
Epoch 132, Training loss 0.5086856484413147, Test loss 0.5053436160087585
Epoch 133, Training loss 0.5084059238433838, Test loss 0.5050488710403442
Epoch 134, Training loss 0.5081287622451782, Test loss 0.5047571659088135
Epoch 135, Training loss 0.507854163646698, Test loss 0.5044683814048767
Epoch 136, Training loss 0.5075821280479431, Test loss 0.5041826367378235
Epoch 137, Training loss 0.5073125958442688, Test loss 0.5038997530937195
Epoch 138, Training loss 0.507045567035675, Test loss 0.5036197304725647
Epoch 139, Training loss 0.5067809820175171, Test loss 0.5033425688743591
Epoch 140, Training loss 0.5065187811851501, Test loss 0.503068208694458
Epoch 141, Training loss 0.506259024143219, Test loss 0.5027965307235718
Epoch 142, Training loss 0.5060016512870789, Test loss 0.50252765417099
Epoch 143, Training loss 0.5057466626167297, Test loss 0.5022614598274231
Epoch 144, Training loss 0.5054940581321716, Test loss 0.5019978880882263
Epoch 145, Training loss 0.505243718624115, Test loss 0.5017369985580444
Epoch 146, Training loss 0.5049957036972046, Test loss 0.5014786720275879
Epoch 147, Training loss 0.5047499537467957, Test loss 0.5012230277061462
Epoch 148, Training loss 0.5045064687728882, Test loss 0.5009698271751404
Epoch 149, Training loss 0.5042652487754822, Test loss 0.5007191896438599
Epoch 150, Training loss 0.5040262937545776, Test loss 0.5004710555076599
Epoch 151, Training loss 0.5037896037101746, Test loss 0.5002254247665405
Epoch 152, Training loss 0.5035551190376282, Test loss 0.4999822974205017
```

```
Epoch 153, Training loss 0.5033227801322937, Test loss 0.4997415244579315
Epoch 154, Training loss 0.5030926465988159, Test loss 0.4995032846927643
Epoch 155, Training loss 0.5028647184371948, Test loss 0.49926742911338806
Epoch 156, Training loss 0.5026388764381409, Test loss 0.49903401732444763
Epoch 157, Training loss 0.5024152398109436, Test loss 0.49880290031433105
Epoch 158, Training loss 0.5021937489509583, Test loss 0.49857422709465027
Epoch 159, Training loss 0.5019744038581848, Test loss 0.4983478784561157
Epoch 160, Training loss 0.5017570853233337, Test loss 0.4981238543987274
Epoch 161, Training loss 0.5015419125556946, Test loss 0.49790212512016296
Epoch 162, Training loss 0.5013288855552673, Test loss 0.4976828098297119
Epoch 163, Training loss 0.5011178851127625, Test loss 0.49746567010879517
Epoch 164, Training loss 0.5009089708328247, Test loss 0.49725088477134705
Epoch 165, Training loss 0.5007021427154541, Test loss 0.4970383942127228
Epoch 166, Training loss 0.5004973411560059, Test loss 0.4968281090259552
Epoch 167, Training loss 0.5002946853637695, Test loss 0.4966200888156891
Epoch 168, Training loss 0.5000938773155212, Test loss 0.49641430377960205
Epoch 169, Training loss 0.49989527463912964, Test loss 0.4962106943130493
Epoch 170, Training loss 0.49969857931137085, Test loss 0.49600932002067566
Epoch 171, Training loss 0.49950388073921204, Test loss 0.49581006169319153
Epoch 172, Training loss 0.4993112087249756, Test loss 0.49561309814453125
Epoch 173, Training loss 0.4991205632686615, Test loss 0.4954181909561157
Epoch 174, Training loss 0.4989318251609802, Test loss 0.4952254891395569
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Epoch 1005, Training loss 0.45770594477653503, Test loss 0.47192972898483276
Epoch 1006, Training loss 0.4576694071292877, Test loss 0.47192901372909546
Epoch 1007, Training loss 0.4576328694820404, Test loss 0.471928209066391
Epoch 1008, Training loss 0.45759642124176025, Test loss 0.4719274938106537
Epoch 1009, Training loss 0.4575600326061249, Test loss 0.4719267785549164
Epoch 1010, Training loss 0.4575236141681671, Test loss 0.47192612290382385
Epoch 1011, Training loss 0.4574873149394989, Test loss 0.4719254672527313
Epoch 1012, Training loss 0.4574510455131531, Test loss 0.471924751996994
Epoch 1013, Training loss 0.45741480588912964, Test loss 0.4719240963459015
Epoch 1014, Training loss 0.457378625869751, Test loss 0.47192347049713135
Epoch 1015, Training loss 0.4573425352573395, Test loss 0.4719228148460388
Epoch 1016, Training loss 0.4573064148426056, Test loss 0.47192221879959106
```

Epoch 1017, Training loss 0.45727041363716125, Test loss 0.4719215929508209
Epoch 1018, Training loss 0.4572344124317169, Test loss 0.47192099690437317
Epoch 1019, Training loss 0.45719847083091736, Test loss 0.471920371055603
Epoch 1020, Training loss 0.4571625888347626, Test loss 0.47191986441612244
Epoch 1021, Training loss 0.45712676644325256, Test loss 0.4719192683696747
Epoch 1022, Training loss 0.45709097385406494, Test loss 0.4719187617301941
Epoch 1023, Training loss 0.4570552110671997, Test loss 0.4719181954860687
Epoch 1024, Training loss 0.45701947808265686, Test loss 0.47191768884658813
Epoch 1025, Training loss 0.45698386430740356, Test loss 0.47191712260246277
Epoch 1026, Training loss 0.45694825053215027, Test loss 0.4719166159629822
Epoch 1027, Training loss 0.45691266655921936, Test loss 0.4719161093235016
Epoch 1028, Training loss 0.4568771421909332, Test loss 0.47191569209098816
Epoch 1029, Training loss 0.45684167742729187, Test loss 0.47191518545150757
Epoch 1030, Training loss 0.4568062722682953, Test loss 0.47191473841667175
Epoch 1031, Training loss 0.4567708969116211, Test loss 0.47191429138183594
Epoch 1032, Training loss 0.4567355215549469, Test loss 0.47191381454467773
Epoch 1033, Training loss 0.4567002058029175, Test loss 0.4719134271144867
Epoch 1034, Training loss 0.4566649794578552, Test loss 0.4719129502773285
Epoch 1035, Training loss 0.45662981271743774, Test loss 0.47191256284713745
Epoch 1036, Training loss 0.4565946161746979, Test loss 0.471912145614624
Epoch 1037, Training loss 0.45655953884124756, Test loss 0.4719117283821106
Epoch 1038, Training loss 0.45652449131011963, Test loss 0.47191137075424194
Epoch 1039, Training loss 0.4564894139766693, Test loss 0.4719109833240509
Epoch 1040, Training loss 0.45645445585250854, Test loss 0.47191062569618225
Epoch 1041, Training loss 0.4564194977283478, Test loss 0.4719102084636688
Epoch 1042, Training loss 0.4563846290111542, Test loss 0.47190991044044495
Epoch 1043, Training loss 0.45634979009628296, Test loss 0.4719095528125763
Epoch 1044, Training loss 0.4563150107860565, Test loss 0.47190919518470764
Epoch 1045, Training loss 0.45628026127815247, Test loss 0.47190889716148376
Epoch 1046, Training loss 0.4562455415725708, Test loss 0.4719085693359375
Epoch 1047, Training loss 0.4562108814716339, Test loss 0.47190821170806885
Epoch 1048, Training loss 0.456176221370697, Test loss 0.47190791368484497
Epoch 1049, Training loss 0.4561416506767273, Test loss 0.4719076156616211
Epoch 1050, Training loss 0.45610713958740234, Test loss 0.4719073176383972
Epoch 1051, Training loss 0.4560726583003998, Test loss 0.4719070494174957
Epoch 1052, Training loss 0.4560382068157196, Test loss 0.47190675139427185
Epoch 1053, Training loss 0.4560038149356842, Test loss 0.471906453371048
Epoch 1054, Training loss 0.4559694528579712, Test loss 0.4719061851501465
Epoch 1055, Training loss 0.45593515038490295, Test loss 0.4719059467315674
Epoch 1056, Training loss 0.4559008777141571, Test loss 0.4719056785106659
Epoch 1057, Training loss 0.45586660504341125, Test loss 0.4719054400920868
Epoch 1058, Training loss 0.45583242177963257, Test loss 0.4719052314758301
Epoch 1059, Training loss 0.45579829812049866, Test loss 0.4719049632549286
Epoch 1060, Training loss 0.45576414465904236, Test loss 0.4719047248363495
Epoch 1061, Training loss 0.455730140209198, Test loss 0.4719045162200928
Epoch 1062, Training loss 0.45569610595703125, Test loss 0.47190430760383606
Epoch 1063, Training loss 0.4556621313095093, Test loss 0.47190409898757935
Epoch 1064, Training loss 0.4556282162666321, Test loss 0.47190389037132263

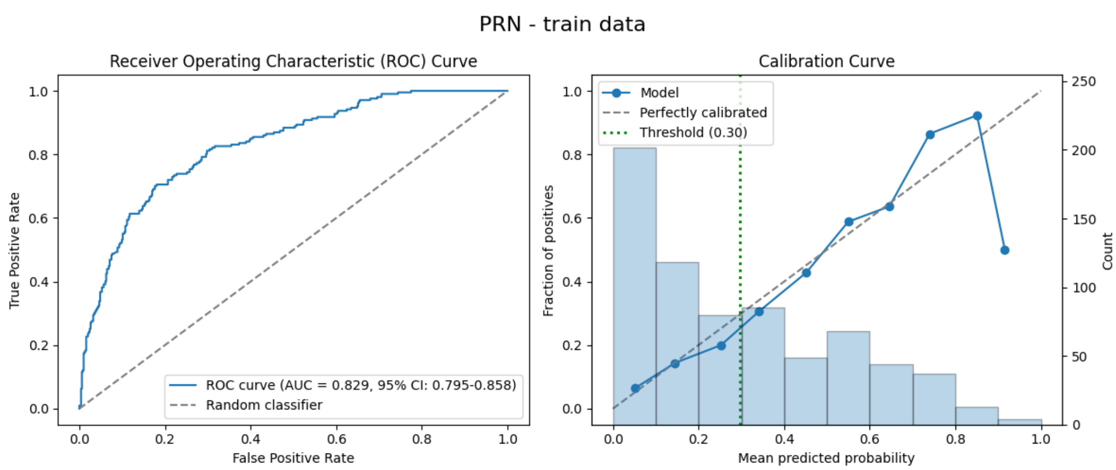
```
Epoch 1065, Training loss 0.4555943012237549, Test loss 0.47190365195274353
Epoch 1066, Training loss 0.45556047558784485, Test loss 0.4719035029411316
Epoch 1067, Training loss 0.4555266797542572, Test loss 0.4719032943248749
Epoch 1068, Training loss 0.45549288392066956, Test loss 0.47190308570861816
Epoch 1069, Training loss 0.4554591476917267, Test loss 0.4719029366970062
Epoch 1070, Training loss 0.455425500869751, Test loss 0.4719027280807495
Epoch 1071, Training loss 0.45539188385009766, Test loss 0.4719025790691376
Epoch 1072, Training loss 0.45535826683044434, Test loss 0.47190237045288086
Epoch 1073, Training loss 0.4553247094154358, Test loss 0.4719022810459137
Epoch 1074, Training loss 0.45529118180274963, Test loss 0.471902072429657
Epoch 1075, Training loss 0.45525774359703064, Test loss 0.47190192341804504
Epoch 1076, Training loss 0.45522433519363403, Test loss 0.4719017744064331
Epoch 1077, Training loss 0.4551909565925598, Test loss 0.47190165519714355
Epoch 1078, Training loss 0.455157607793808, Test loss 0.47190144658088684
Epoch 1079, Training loss 0.45512428879737854, Test loss 0.4719013571739197
Epoch 1080, Training loss 0.45509102940559387, Test loss 0.4719012677669525
Epoch 1081, Training loss 0.45505785942077637, Test loss 0.4719011187553406
Epoch 1082, Training loss 0.4550246596336365, Test loss 0.47190093994140625
Epoch 1083, Training loss 0.45499154925346375, Test loss 0.4719008505344391
Epoch 1084, Training loss 0.45495840907096863, Test loss 0.4719007611274719
Epoch 1085, Training loss 0.4549253582954407, Test loss 0.4719006419181824
Epoch 1086, Training loss 0.4548923671245575, Test loss 0.47190049290657043
Epoch 1087, Training loss 0.45485934615135193, Test loss 0.47190040349960327
Epoch 1088, Training loss 0.4548264741897583, Test loss 0.4719003438949585
Epoch 1089, Training loss 0.4547935724258423, Test loss 0.47190019488334656
Epoch 1090, Training loss 0.45476073026657104, Test loss 0.471900075674057
Epoch 1091, Training loss 0.4547279179096222, Test loss 0.4719000458717346
Epoch 1092, Training loss 0.4546951651573181, Test loss 0.47189992666244507
Epoch 1093, Training loss 0.45466241240501404, Test loss 0.4718998372554779
Epoch 1094, Training loss 0.4546297788619995, Test loss 0.47189971804618835
Epoch 1095, Training loss 0.4545970857143402, Test loss 0.4718996286392212
Epoch 1096, Training loss 0.4545644521713257, Test loss 0.4718996286392212
Epoch 1097, Training loss 0.4545319378376007, Test loss 0.47189947962760925
Epoch 1098, Training loss 0.45449936389923096, Test loss 0.4718994200229645
Epoch 1099, Training loss 0.45446690917015076, Test loss 0.4718993306159973
Epoch 1100, Training loss 0.45443445444107056, Test loss 0.47189921140670776
Epoch 1101, Training loss 0.45440205931663513, Test loss 0.4718991816043854
Epoch 1102, Training loss 0.4543696641921997, Test loss 0.4718991219997406
Epoch 1103, Training loss 0.45433735847473145, Test loss 0.4718990623950958
Epoch 1104, Training loss 0.45430511236190796, Test loss 0.47189903259277344
Epoch 1105, Training loss 0.45427289605140686, Test loss 0.4718989133834839
Epoch 1106, Training loss 0.45424067974090576, Test loss 0.4718988537788391
Epoch 1107, Training loss 0.45420849323272705, Test loss 0.4718988239765167
Epoch 1108, Training loss 0.4541763663291931, Test loss 0.4718987047672272
Epoch 1109, Training loss 0.45414426922798157, Test loss 0.4718986749649048
Epoch 1110, Training loss 0.4541122317314148, Test loss 0.4718986749649048
Epoch 1111, Training loss 0.4540802240371704, Test loss 0.47189861536026
Epoch 1112, Training loss 0.4540482759475708, Test loss 0.47189849615097046
```

```
Epoch 1113, Training loss 0.4540163576602936, Test loss 0.47189849615097046
Epoch 1114, Training loss 0.45398446917533875, Test loss 0.4718984067440033
Epoch 1115, Training loss 0.4539526402950287, Test loss 0.4718984067440033
Epoch 1116, Training loss 0.45392081141471863, Test loss 0.47189831733703613
Epoch 1117, Training loss 0.45388907194137573, Test loss 0.47189825773239136
Epoch 1118, Training loss 0.45385733246803284, Test loss 0.47189825773239136
Epoch 1119, Training loss 0.45382562279701233, Test loss 0.4718981981277466
Epoch 1120, Training loss 0.4537939727306366, Test loss 0.4718981981277466
Epoch 1121, Training loss 0.45376238226890564, Test loss 0.4718981087207794
Epoch 1122, Training loss 0.45373082160949707, Test loss 0.4718981087207794
Epoch 1123, Training loss 0.4536992907524109, Test loss 0.4718980491161346
Epoch 1124, Training loss 0.4536677598953247, Test loss 0.47189798951148987
Epoch 1125, Training loss 0.4536362886428833, Test loss 0.4718979597091675
Epoch 1126, Training loss 0.45360490679740906, Test loss 0.4718979001045227
Epoch 1127, Training loss 0.4535735547542572, Test loss 0.47189784049987793
Epoch 1128, Training loss 0.45354220271110535, Test loss 0.47189784049987793
Epoch 1129, Training loss 0.45351091027259827, Test loss 0.47189781069755554
Epoch 1130, Training loss 0.4534796476364136, Test loss 0.47189775109291077
Epoch 1131, Training loss 0.45344844460487366, Test loss 0.471897691488266
Epoch 1132, Training loss 0.45341721177101135, Test loss 0.471897691488266
Epoch 1133, Training loss 0.4533860981464386, Test loss 0.4718976318836212
Epoch 1134, Training loss 0.45335501432418823, Test loss 0.47189760208129883
Epoch 1135, Training loss 0.45332396030426025, Test loss 0.47189754247665405
Epoch 1136, Training loss 0.4532929360866546, Test loss 0.4718974828720093
Epoch 1137, Training loss 0.45326194167137146, Test loss 0.4718974828720093
Epoch 1138, Training loss 0.45323097705841064, Test loss 0.4718974530696869
Epoch 1139, Training loss 0.4532000720500946, Test loss 0.4718973934650421
Epoch 1140, Training loss 0.45316919684410095, Test loss 0.4718973934650421
Epoch 1141, Training loss 0.4531383216381073, Test loss 0.47189730405807495
Epoch 1142, Training loss 0.4531075656414032, Test loss 0.47189730405807495
Epoch 1143, Training loss 0.4530768096446991, Test loss 0.4718972444534302
Epoch 1144, Training loss 0.4530460834503174, Test loss 0.4718971848487854
Epoch 1145, Training loss 0.45301538705825806, Test loss 0.4718971252441406
Epoch 1146, Training loss 0.4529847502708435, Test loss 0.47189709544181824
Epoch 1147, Training loss 0.45295417308807373, Test loss 0.47189709544181824
Epoch 1148, Training loss 0.45292356610298157, Test loss 0.4718969762325287
Epoch 1149, Training loss 0.45289304852485657, Test loss 0.4718969762325287
Epoch 1150, Training loss 0.45286259055137634, Test loss 0.4718968868255615
Epoch 1151, Training loss 0.45283210277557373, Test loss 0.4718968868255615
Epoch 1152, Training loss 0.45280173420906067, Test loss 0.47189682722091675
Epoch 1153, Training loss 0.4527713358402252, Test loss 0.471896767616272
Early stopping! Epoch 1153
Best model at epoch 1053
```

[321]:
```
# Evaluate PRN on training data
y_pred_train_prn = partial_response_network.predict(X_train_tensor,␣
↪device=device)
```

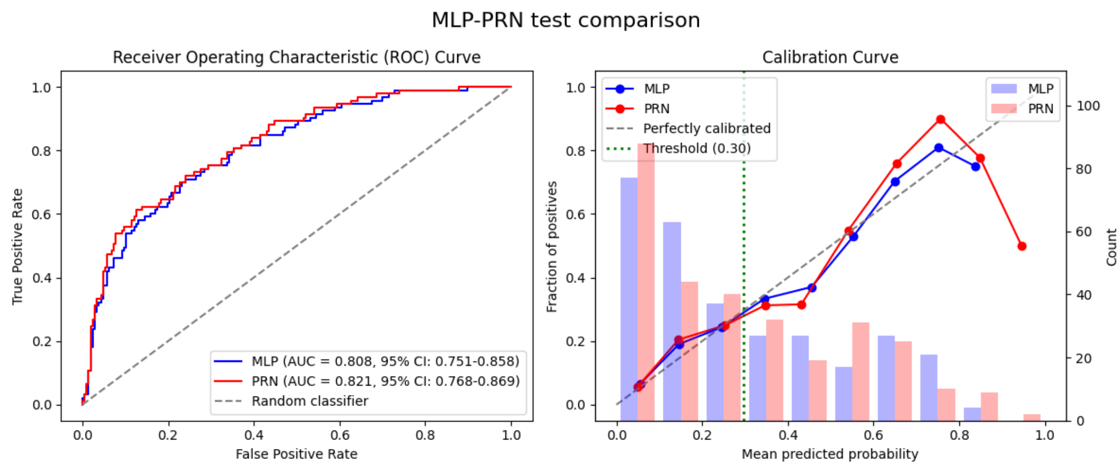
```python
results_prn_train = evaluate_model_performance(y_train, y_pred_train_prn, y_train, title="PRN - train data")

# Evaluate PRN on test data and compare to MLP (blackbox)
y_pred_test_prn = partial_response_network.predict(X_test_tensor, device=device)

results_prn_mlp_test = compare_model_performance(y_test, y_pred_test_blackbox, y_pred_test_prn, y_train=y_train, model_names=("MLP","PRN"), title="MLP-PRN test comparison")
```

```
---- Model Performance Metrics PRN - train data ----
Threshold (prevalence) set to: 0.296
AUROC:          0.829 (95% CI: 0.795-0.858) (Area Under the Receiver Operating
Characteristic curve)
Accuracy:       0.736 (Proportion of correct predictions)
Precision:      0.536 (Proportion of true positives among positive predictions,
aka PPV)
Recall:         0.792 (Proportion of true positives among actual positives, aka
sensitivity)
F1 Score:       0.639 (Harmonic mean of precision and recall)
--------------------------------------------
```




```
-------- Model Performance Comparison --------
Threshold (prevalence) set to: 0.296

MLP:
AUROC:          0.808 (95% CI: 0.751-0.858)
Accuracy:       0.723
Precision:      0.540
```

```
Recall:          0.720
F1 Score:        0.618

PRN:
AUROC:           0.821 (95% CI: 0.768-0.869)
Accuracy:        0.717
Precision:       0.531
Recall:          0.742
F1 Score:        0.619
------------------------------------------------
```



1.9 9. LASSO on PRN

We perform LASSO again, this time on the Partial Response Network to isolate the most important features and interactions.

[322]:
```
partial_responses_params_prn = {
    'x_train': X_train_tensor,
    'method': partial_response_method,
    'device': device,
    'batch_size': 256,
    'max_workers': max_workers
}

partial_responses_train_prn = partial_responses(X_train_tensor,␣
 ↪partial_response_network, **partial_responses_params_prn)
partial_responses_test_prn = partial_responses(X_test_tensor,␣
 ↪partial_response_network, **partial_responses_params_prn)
```

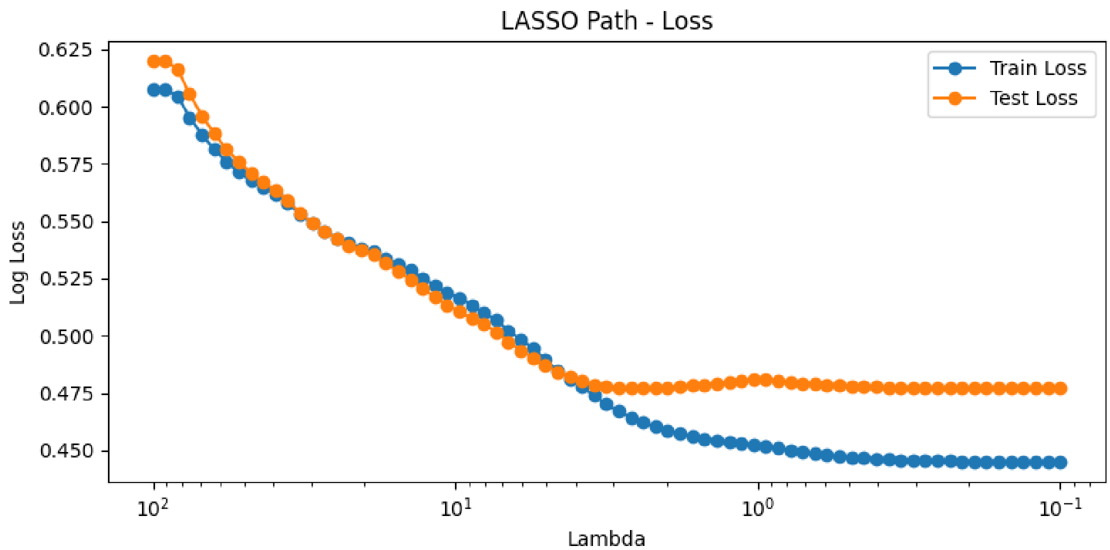
```
Main compute device: cuda
Max threads: 1
Batch size: 256
```

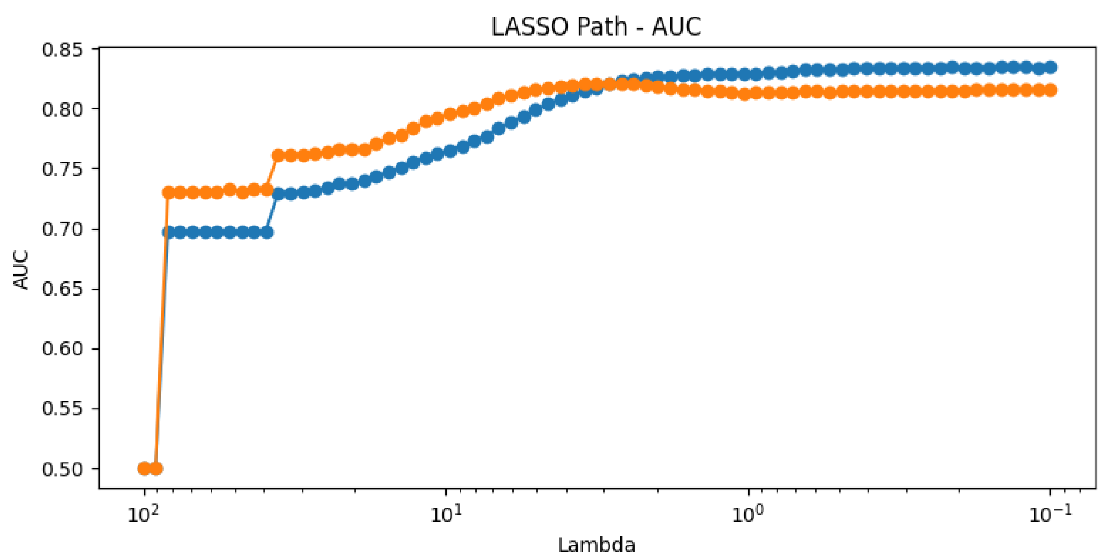
```
Univariate 0,   (cuda)
Univariate 1,   (cuda)
Univariate 2,   (cuda)
Univariate 3,   (cuda)
Univariate 4,   (cuda)
Univariate 5,   (cuda)
Univariate 6,   (cuda)
Univariate 7,   (cuda)
Univariate 8,   (cuda)
Univariate 9,   (cuda)
Bivariate 0,1,  (cuda)
Bivariate 0,2,  (cuda)
Bivariate 0,3,  (cuda)
Bivariate 0,4,  (cuda)
Bivariate 0,5,  (cuda)
Bivariate 0,6,  (cuda)
Bivariate 0,7,  (cuda)
Bivariate 0,8,  (cuda)
Bivariate 0,9,  (cuda)
Bivariate 1,2,  (cuda)
Bivariate 1,3,  (cuda)
Bivariate 1,4,  (cuda)
Bivariate 1,5,  (cuda)
Bivariate 1,6,  (cuda)
Bivariate 1,7,  (cuda)
Bivariate 1,8,  (cuda)
Bivariate 1,9,  (cuda)
Bivariate 2,3,  (cuda)
Bivariate 2,4,  (cuda)
Bivariate 2,5,  (cuda)
Bivariate 2,6,  (cuda)
Bivariate 2,7,  (cuda)
Bivariate 2,8,  (cuda)
Bivariate 2,9,  (cuda)
Bivariate 3,4,  (cuda)
Bivariate 3,5,  (cuda)
Bivariate 3,6,  (cuda)
Bivariate 3,7,  (cuda)
Bivariate 3,8,  (cuda)
Bivariate 3,9,  (cuda)
Bivariate 4,5,  (cuda)
Bivariate 4,6,  (cuda)
Bivariate 4,7,  (cuda)
Bivariate 4,8,  (cuda)
Bivariate 4,9,  (cuda)
Bivariate 5,6,  (cuda)
Bivariate 5,7,  (cuda)
Bivariate 5,8,  (cuda)
```

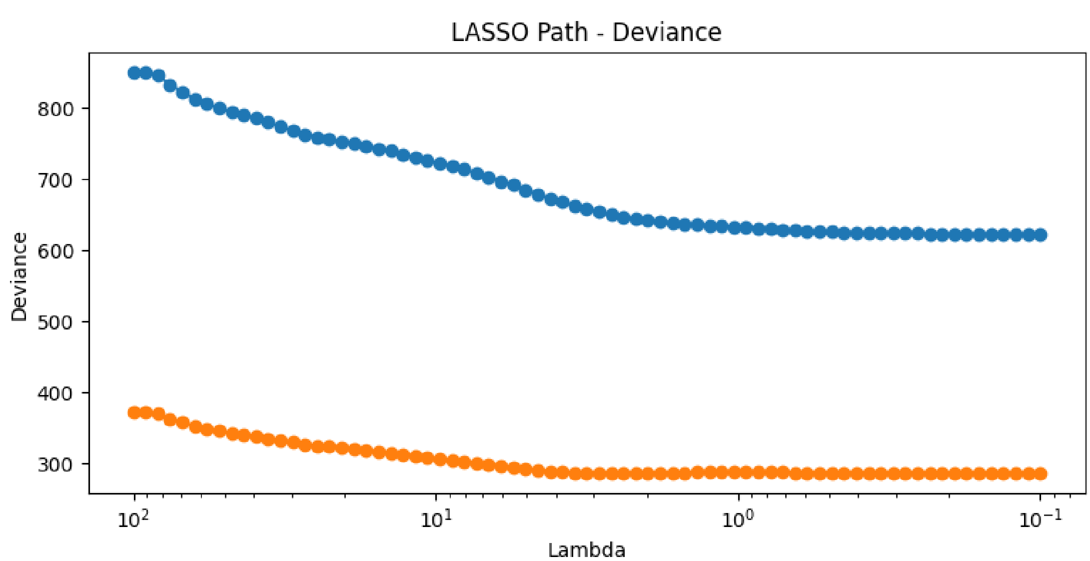
```
Bivariate 5,9,  (cuda)
Bivariate 6,7,  (cuda)
Bivariate 6,8,  (cuda)
Bivariate 6,9,  (cuda)
Bivariate 7,8,  (cuda)
Bivariate 7,9,  (cuda)
Bivariate 8,9,  (cuda)
Main compute device: cuda
Max threads: 1
Batch size: 256
Univariate 0,   (cuda)
Univariate 1,   (cuda)
Univariate 2,   (cuda)
Univariate 3,   (cuda)
Univariate 4,   (cuda)
Univariate 5,   (cuda)
Univariate 6,   (cuda)
Univariate 7,   (cuda)
Univariate 8,   (cuda)
Univariate 9,   (cuda)
Bivariate 0,1,  (cuda)
Bivariate 0,2,  (cuda)
Bivariate 0,3,  (cuda)
Bivariate 0,4,  (cuda)
Bivariate 0,5,  (cuda)
Bivariate 0,6,  (cuda)
Bivariate 0,7,  (cuda)
Bivariate 0,8,  (cuda)
Bivariate 0,9,  (cuda)
Bivariate 1,2,  (cuda)
Bivariate 1,3,  (cuda)
Bivariate 1,4,  (cuda)
Bivariate 1,5,  (cuda)
Bivariate 1,6,  (cuda)
Bivariate 1,7,  (cuda)
Bivariate 1,8,  (cuda)
Bivariate 1,9,  (cuda)
Bivariate 2,3,  (cuda)
Bivariate 2,4,  (cuda)
Bivariate 2,5,  (cuda)
Bivariate 2,6,  (cuda)
Bivariate 2,7,  (cuda)
Bivariate 2,8,  (cuda)
Bivariate 2,9,  (cuda)
Bivariate 3,4,  (cuda)
Bivariate 3,5,  (cuda)
Bivariate 3,6,  (cuda)
Bivariate 3,7,  (cuda)
```

```
Bivariate 3,8,  (cuda)
Bivariate 3,9,  (cuda)
Bivariate 4,5,  (cuda)
Bivariate 4,6,  (cuda)
Bivariate 4,7,  (cuda)
Bivariate 4,8,  (cuda)
Bivariate 4,9,  (cuda)
Bivariate 5,6,  (cuda)
Bivariate 5,7,  (cuda)
Bivariate 5,8,  (cuda)
Bivariate 5,9,  (cuda)
Bivariate 6,7,  (cuda)
Bivariate 6,8,  (cuda)
Bivariate 6,9,  (cuda)
Bivariate 7,8,  (cuda)
Bivariate 7,9,  (cuda)
Bivariate 8,9,  (cuda)
```

[323]:
```
lasso_results_prn = lasso(
    partial_responses_train_prn,
    partial_responses_test_prn,
    y_train,
    y_test,
    feature_names=feature_names,
    nlambda=75,
    min_lambda=0.1,
    max_lambda=100,
    batch_size=2,
    seed=random_seed
)
```


LASSO Path - Loss
Train Loss
Test Loss
Log Loss
0.625
0.600
0.575
0.550
0.525
0.500
0.475
0.450
10^2
10^1
10^0
10^{-1}
Lambda

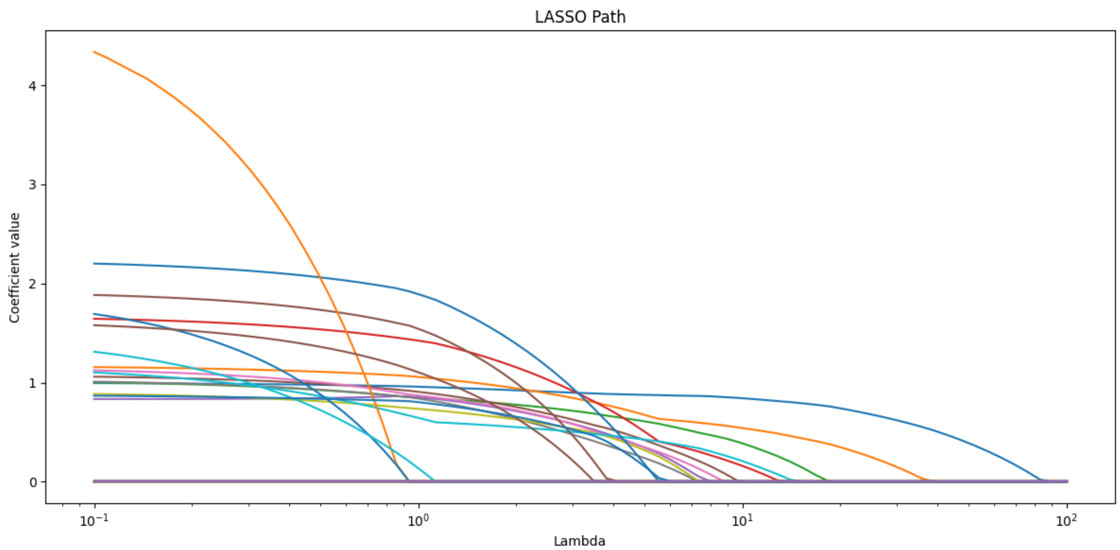

LASSO Path - AUC
AUC
0.85
0.80
0.75
0.70
0.65
0.60
0.55
0.50
10^2
10^1
10^0
10^{-1}
Lambda


LASSO Path - Deviance
Deviance
800
700
600
500
400
300
10^2
10^1
10^0
10^{-1}
Lambda

```
<IPython.core.display.HTML object>
```

```
All fits converged successfully.
```

[331]:
```
lasso_results_prn.plot_lambda_path()
```



[332]:
```
lasso_results_prn.select_lambda_max_test_auc()

# Alternatively, manually select lambda as needed
# lasso_results_prn.select_lambda(38)
```

```
Selected lambda index: 38
Selected lambda value: 2.8804
Corresponding test AUC: 0.8210
```

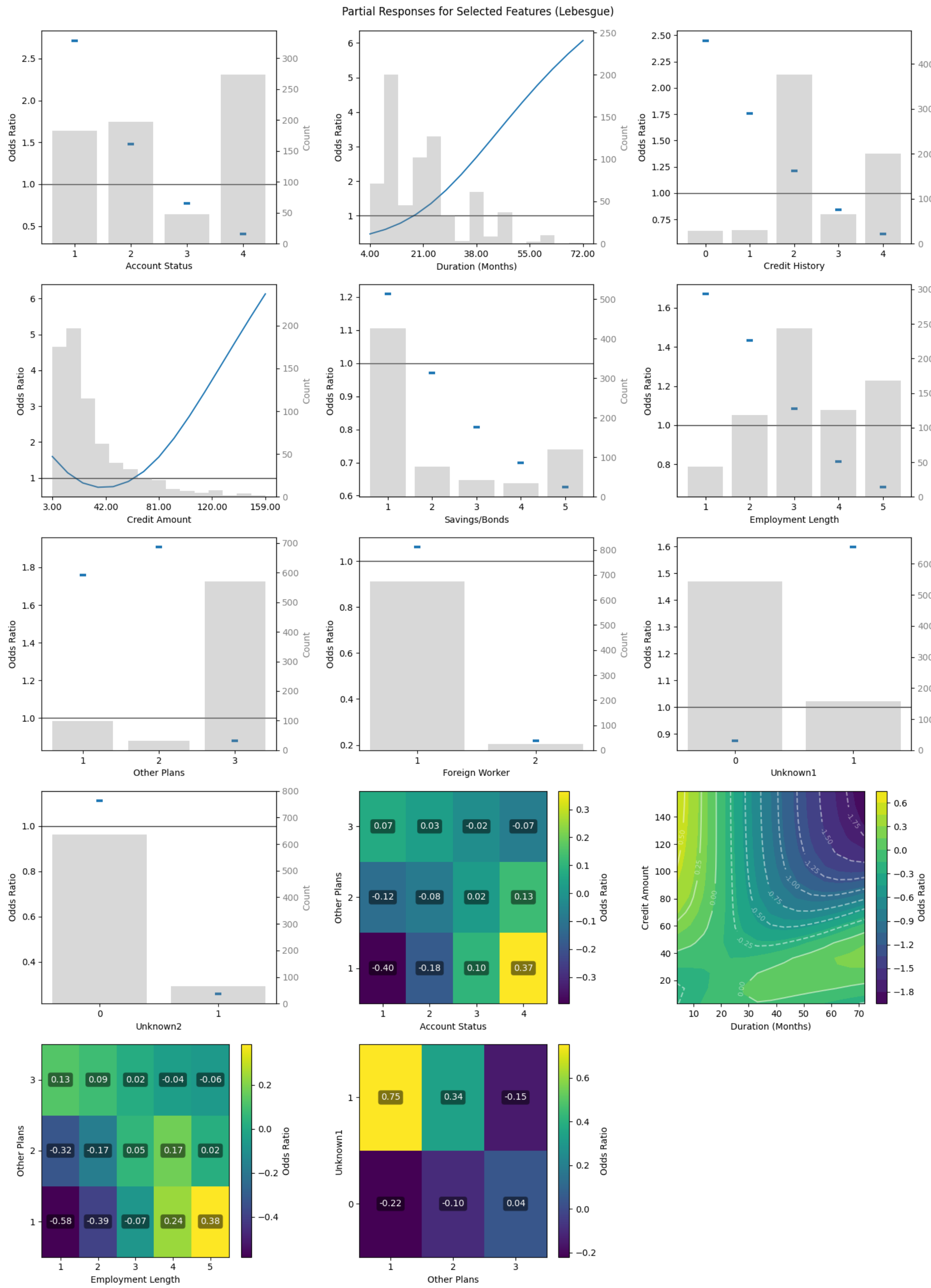
[332]: 38

1.10 10. Generate Final Nomogram

Finally, we generate a nomogram for our PRN, based on the features selected with LASSO. The nomogram provides a visual tool that can be used to understand the model's decisions.

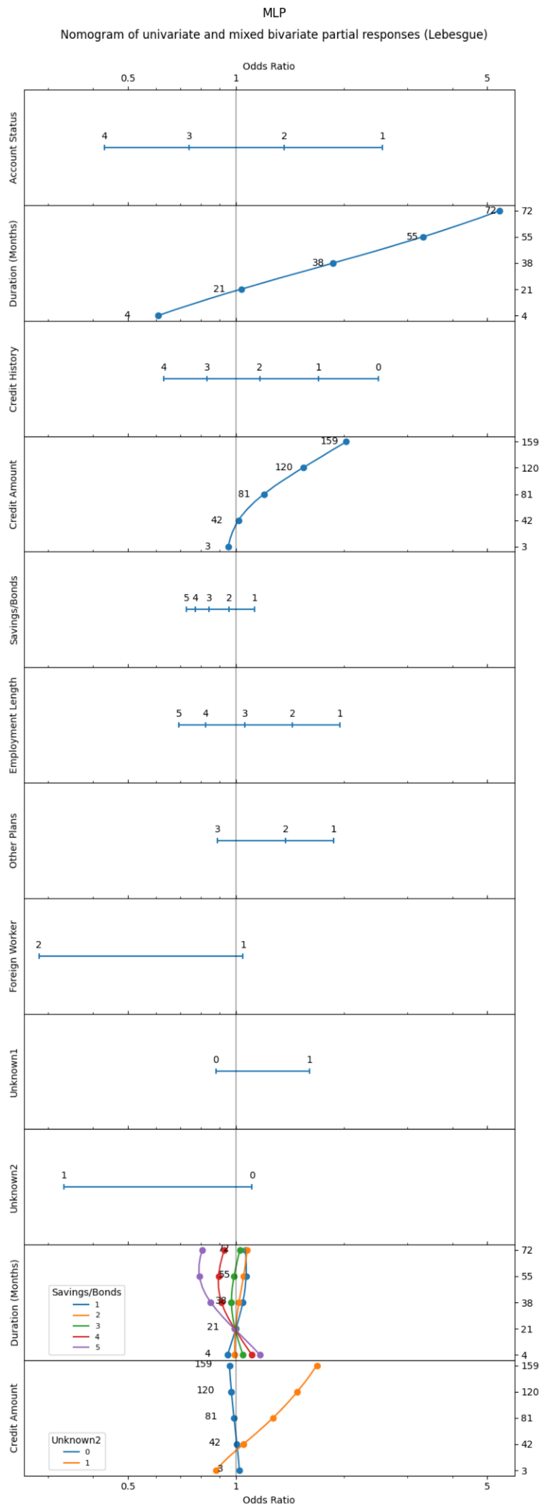
[325]:
```
fig_prn = plot_partial_responses(
    lasso_results_prn,
    X_train_tensor,
    X_train.median().values,
    X_train.std().values,
    partial_response_network,
```

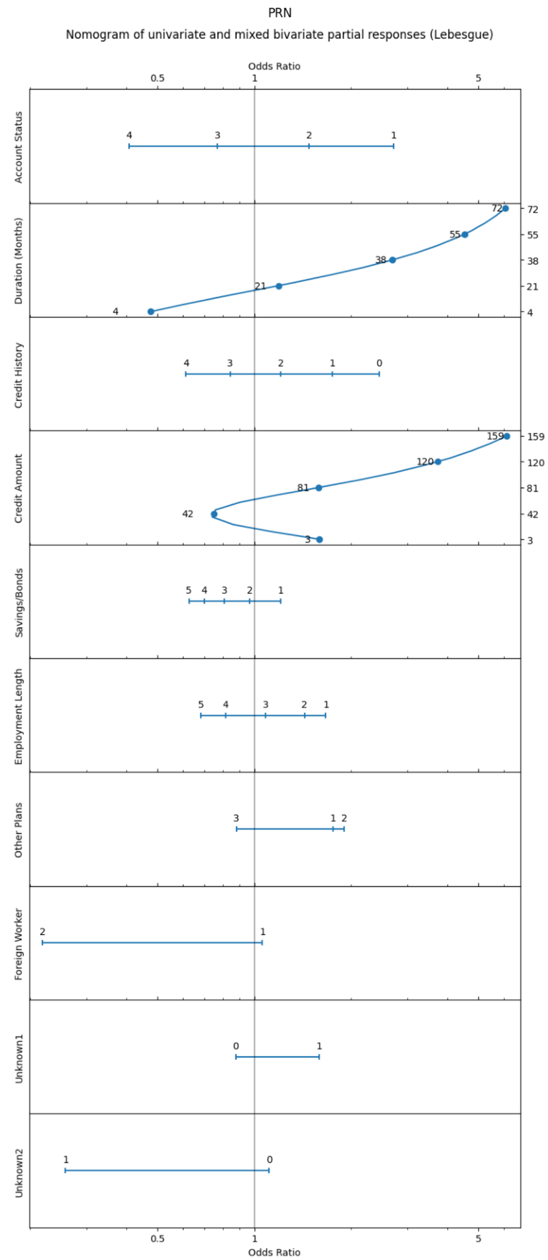
```
    n_steps=15,
    sd_scale=2,
    method=partial_response_method,
    device=device,
    categorical_threshold=15,
    subtract_univariate=True,
    figsize=(15,20),
    show_fig=True,
    return_fig=True,
    use_odds_ratio=True
)
```

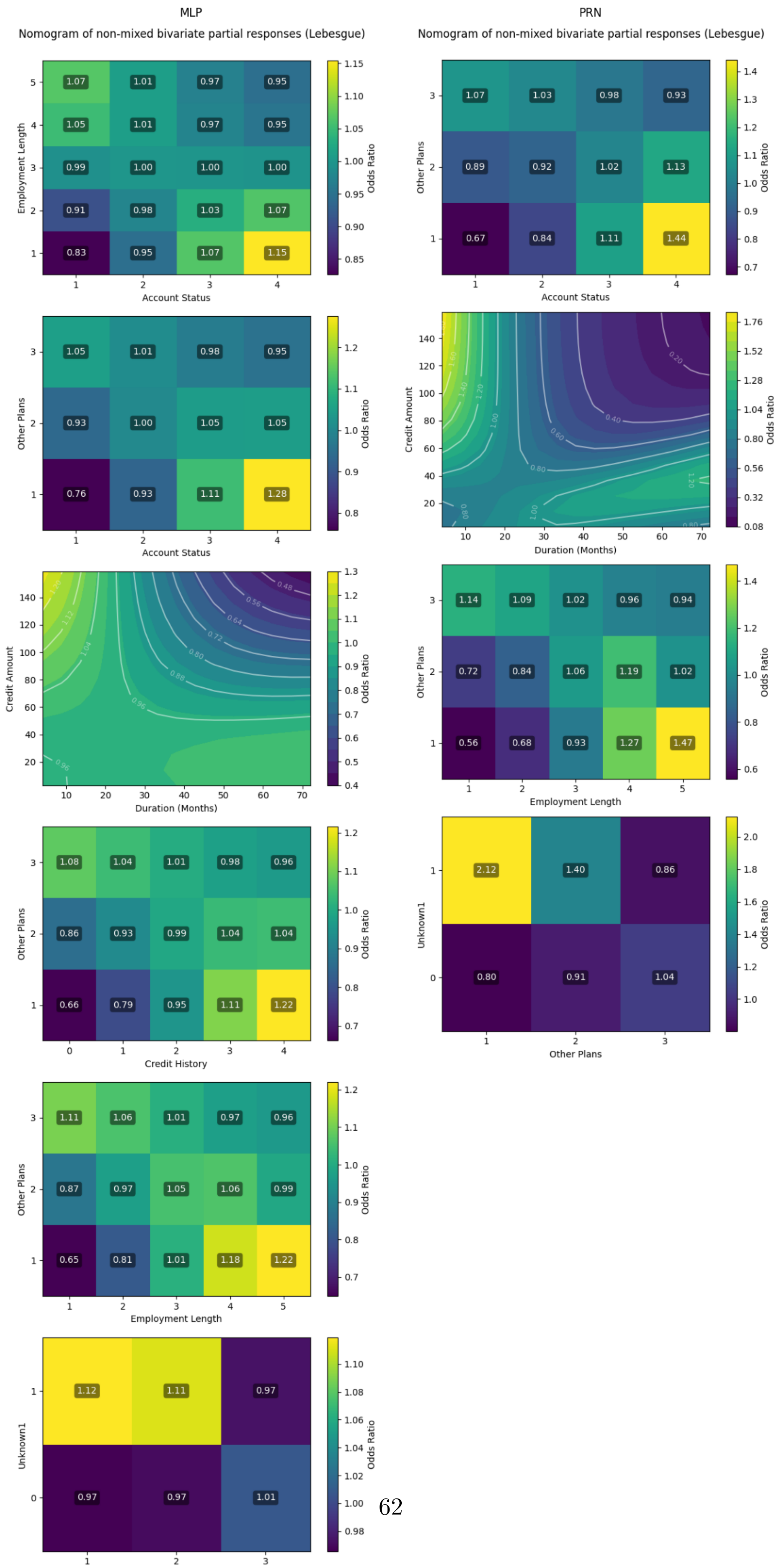

Partial Responses for Selected Features (Lebesgue)
Odds Ratio
Count
Account Status
Duration (Months)
Credit History
Credit Amount
Savings/Bonds
Employment Length
Other Plans
Foreign Worker
Unknown1
Unknown2

[326]:
```
%%capture
# ^ Supress nomogram plot output, will display side-by-side instead

nomogram_params_prn = {
    'n_steps' : 15,
    'sd_scale' : 2,
    'method' : partial_response_method,
    'device' : device,
    'categorical_threshold' : 15,
    'subtract_univariate' : True,
    'show_fig' : False,
    'return_fig' : True,
    'use_odds_ratio' : True,
    'save_csv' : SAVE_METRICS,
    'csv_comment' : "PRN German CC data"
}

nomogram_results_prn = nomogram(
    lasso_results_prn,
    X_train_tensor,
    X_train.median().values,
    X_train.std().values,
    partial_response_network,
    **nomogram_params_prn
)

nomogram_main_prn = nomogram_results_prn[6]
nomogram_non_mixed_prn = nomogram_results_prn[7]
```

[327]:
```
display_nomograms_side_by_side(nomogram_main_mlp, nomogram_non_mixed_mlp,
                               nomogram_main_prn, nomogram_non_mixed_prn)
```


MLP
Nomogram of univariate and mixed bivariate partial responses (Lebesgue)
Odds Ratio
Account Status
Duration (Months)
Credit History
Credit Amount
Savings/Bonds
Employment Length
Other Plans
Foreign Worker
Unknown1
Unknown2
Duration (Months)
Savings/Bonds
Credit Amount
Unknown2
Odds Ratio


PRN
Nomogram of univariate and mixed bivariate partial responses (Lebesgue)
Odds Ratio
Account Status
Duration (Months)
Credit History
Credit Amount
Savings/Bonds
Employment Length
Other Plans
Foreign Worker
Unknown1
Unknown2
Odds Ratio


MLP
Nomogram of non-mixed bivariate partial responses (Lebesgue)
PRN
Nomogram of non-mixed bivariate partial responses (Lebesgue)
Employment Length
Account Status
Odds Ratio
1.07 1.01 0.97 0.95
1.05 1.01 0.97 0.95
0.99 1.00 1.00 1.00
0.91 0.98 1.03 1.07
0.83 0.95 1.07 1.15
Other Plans
Account Status
1.07 1.03 0.98 0.93
0.89 0.92 1.02 1.13
0.67 0.84 1.11 1.44
Other Plans
Account Status
1.05 1.01 0.98 0.95
0.93 1.00 1.05 1.05
0.76 0.93 1.11 1.28
Credit Amount
Duration (Months)
Other Plans
Employment Length
1.14 1.09 1.02 0.96 0.94
0.72 0.84 1.06 1.19 1.02
0.56 0.68 0.93 1.27 1.47
Other Plans
Credit History
1.08 1.04 1.01 0.98 0.96
0.86 0.93 0.99 1.04 1.04
0.66 0.79 0.95 1.11 1.22
Unknown1
Other Plans
2.12 1.40 0.86
0.80 0.91 1.04
Other Plans
Employment Length
1.11 1.06 1.01 0.97 0.96
0.87 0.97 1.05 1.06 0.99
0.65 0.81 1.01 1.18 1.22
Unknown1
Other Plans
1.12 1.11 0.97
0.97 0.97 1.01

1.11 11. Evaluate PRN-LASSO Model

Finally, we evaluate the simplified model (multivariate logisitic regression with LASSO regularization) trained on the partial responses of the PRN, based on the selected lambda.

This notebook demonstrates the PRiSM method, transforming a black box model into an interpretable one without sacrificing predictive performance. This approach is particularly valuable when understanding the model's decision-making process is crucial for building trust and gaining insights.

The final PRN-LASSO model provides both high predictive accuracy and interpretability, allowing us to understand how different features contribute to the predictions.

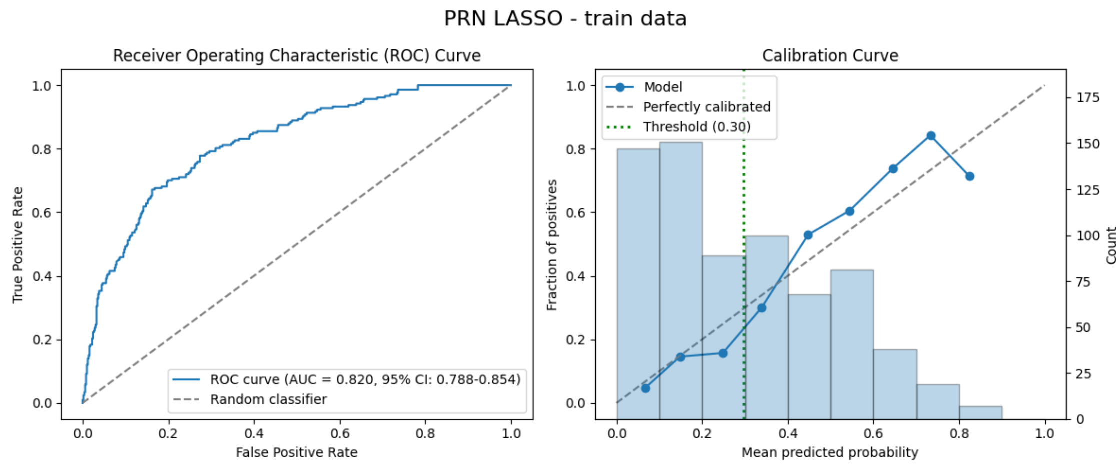
[328]:
```
prn_lasso = lasso_results_prn.get_selected_model()
```

```
Logistic regression model for Lambda index 38 (2.88) selected.
```

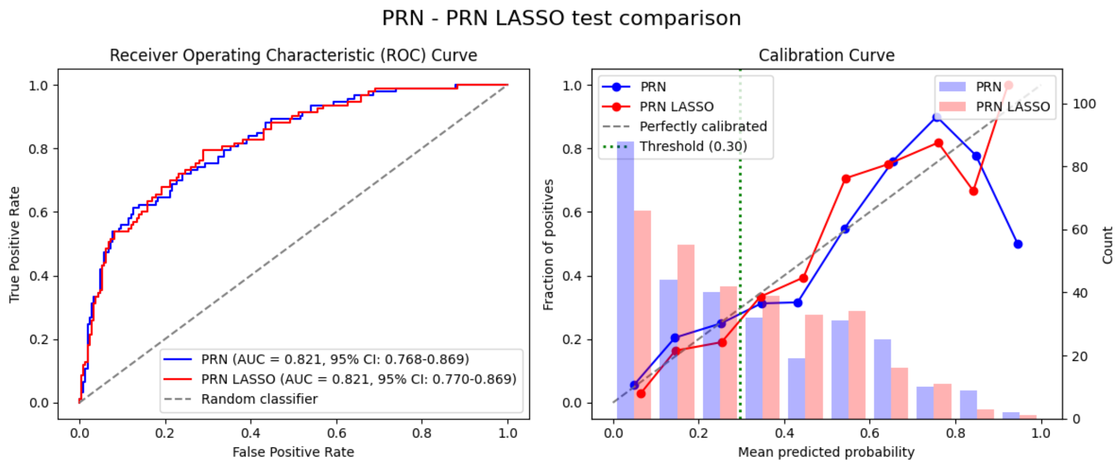
[329]:
```
# Evaluate PRN-LASSO on training data
y_pred_train_prn_lasso = prn_lasso.predict_proba(partial_responses_train_prn)[:
 ↪, 1]

results_prn_lasso_train = evaluate_model_performance(y_train,␣
 ↪y_pred_train_prn_lasso, y_train, title="PRN LASSO - train data")

# Evaluate PRN-LASSO on test data and compare to PRN
y_pred_test_prn_lasso = prn_lasso.predict_proba(partial_responses_test_prn)[:,␣
 ↪1]

results_prn_prn_lasso_test = compare_model_performance(y_test, y_pred_test_prn,␣
 ↪y_pred_test_prn_lasso, y_train=y_train, model_names=("PRN","PRN LASSO"),␣
 ↪title="PRN - PRN LASSO test comparison")

# Compare PRN-LASSO and original MLP (blackbox) model
results_prn_mlp_test = compare_model_performance(y_test, y_pred_test_blackbox,␣
 ↪y_pred_test_prn_lasso, y_train=y_train, model_names=("MLP","PRN LASSO"),␣
 ↪title="MLP - PRN LASSO test comparison")
```

```
---- Model Performance Metrics PRN LASSO - train data ----
Threshold (prevalence) set to: 0.296
AUROC:          0.820 (95% CI: 0.788-0.854) (Area Under the Receiver Operating
Characteristic curve)
Accuracy:       0.721 (Proportion of correct predictions)
Precision:      0.519 (Proportion of true positives among positive predictions,
aka PPV)
Recall:         0.802 (Proportion of true positives among actual positives, aka
sensitivity)
```

```
F1 Score:        0.630 (Harmonic mean of precision and recall)
--------------------------------------------
```




```
-------- Model Performance Comparison --------
Threshold (prevalence) set to: 0.296

PRN:
AUROC:           0.821 (95% CI: 0.768-0.869)
Accuracy:        0.717
Precision:       0.531
Recall:          0.742
F1 Score:        0.619

PRN LASSO:
AUROC:           0.821 (95% CI: 0.770-0.869)
Accuracy:        0.717
Precision:       0.529
Recall:          0.796
F1 Score:        0.635
------------------------------------------------
```




```
-------- Model Performance Comparison --------
Threshold (prevalence) set to: 0.296

MLP:
AUROC:          0.808 (95% CI: 0.755-0.858)
Accuracy:       0.723
Precision:      0.540
Recall:         0.720
F1 Score:       0.618

PRN LASSO:
AUROC:          0.821 (95% CI: 0.765-0.868)
Accuracy:       0.717
Precision:      0.529
Recall:         0.796
F1 Score:       0.635
-------------------------------------------------
```